Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch

Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

CHR Bank Selection Logic

File: chr_bank_selection_logic.kicad_sch

CHR Address Bus 10 to 14

File: chr_address_bus_10_to_14.kicad_sch

CHR Address Bus 15 to 19

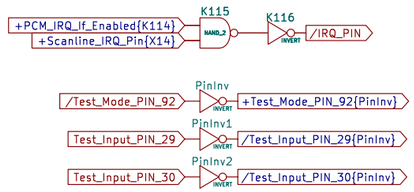
File: chr_address_bus_15_to_19.kicad_sch

Multiplier

File: multiplier.kicad_sch

Artifacts

File: artifacts.kicad_sch



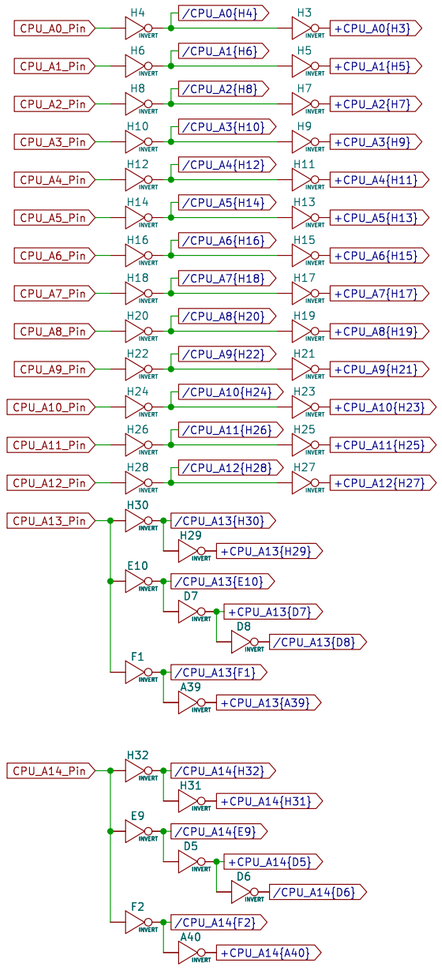
Sheet: /

File: MMC5.kicad_sch

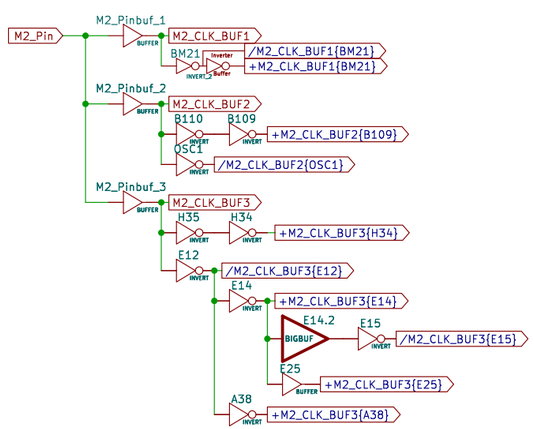
Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/26

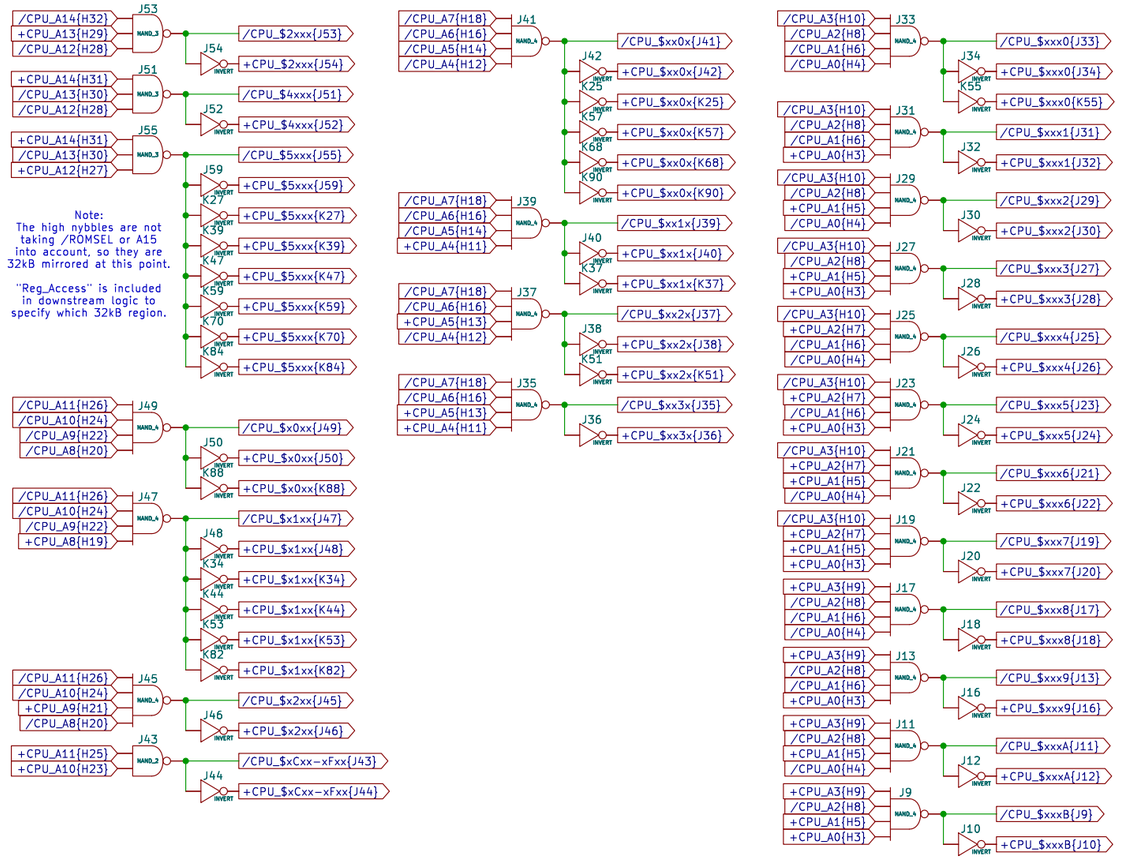
CPU Address Bus Buffering



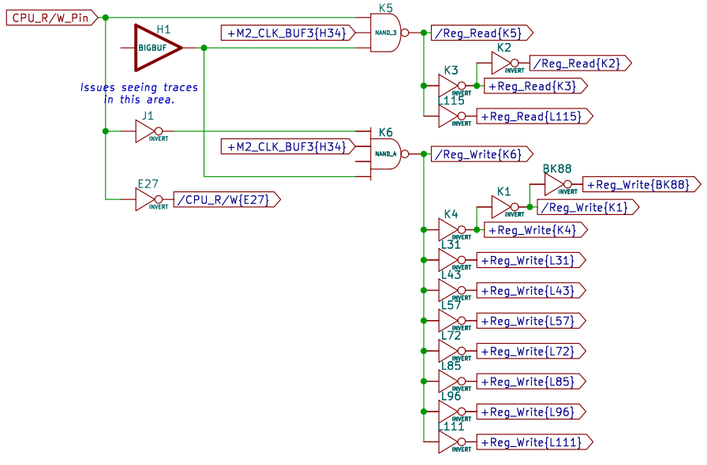
M2 Clock Buffering



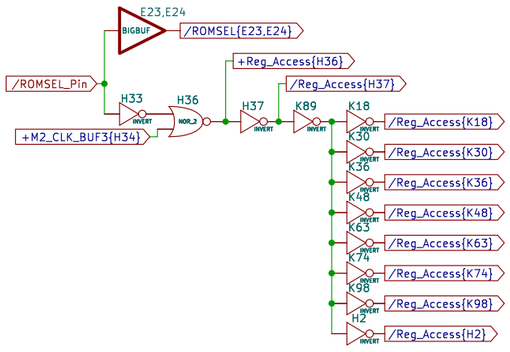
CPU Address Bus Nybble Decoding



CPU R/W, Register Read/Write



/ROMSEL, Register Access

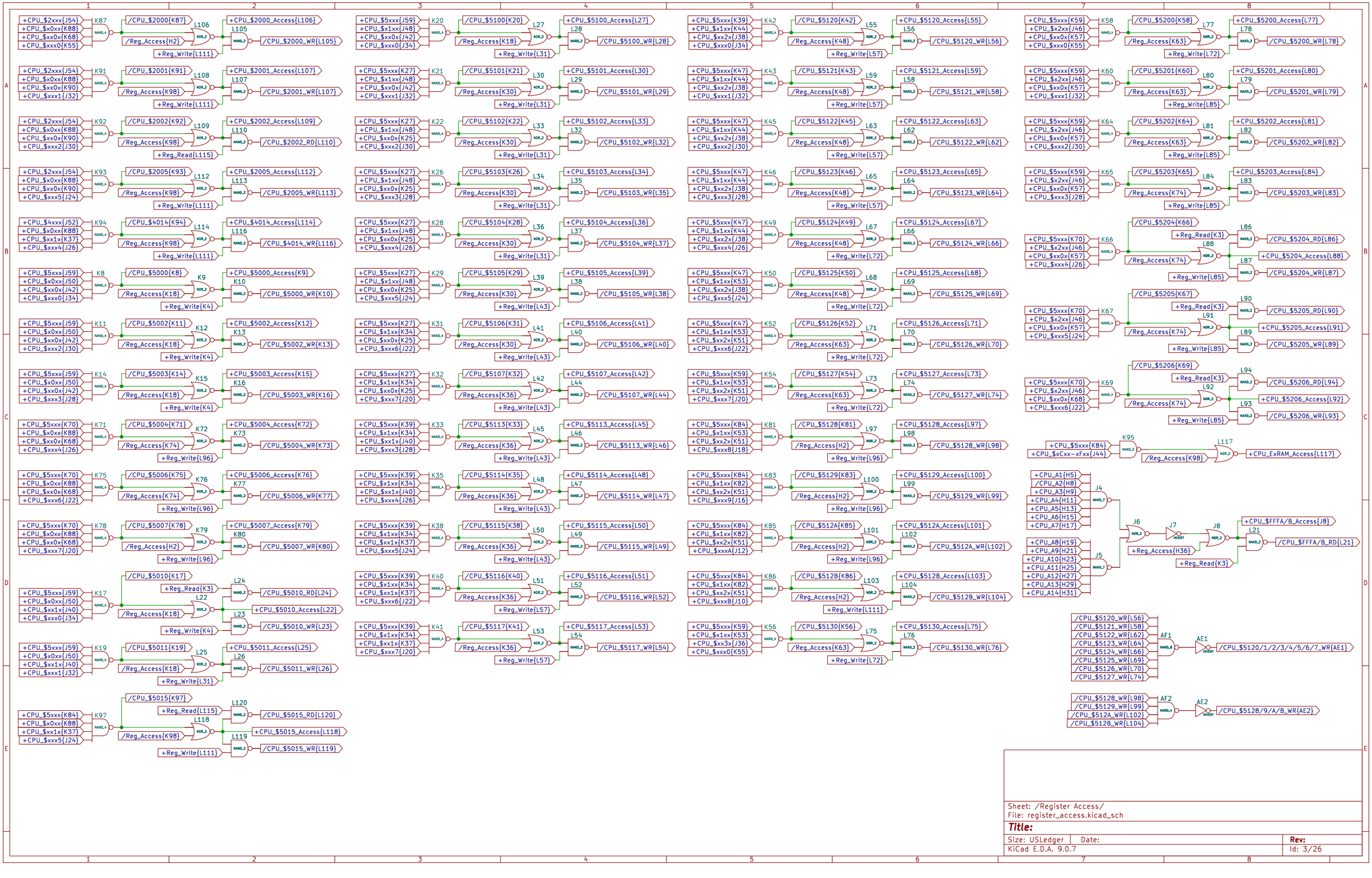


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File: cpuAddressBus.kicad_sch

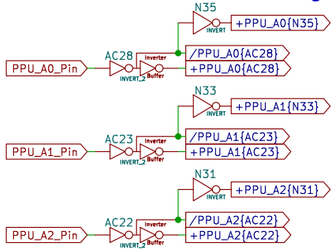
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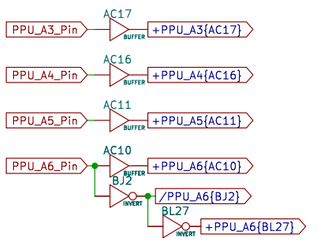
Size: USLedger | Date: | **Rev:**

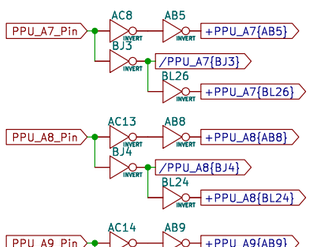
KiCad E.D.A. 9.0.7 | Id: 2/26

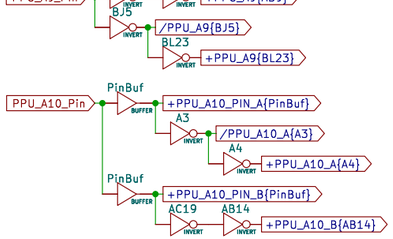

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+CPU_$5xxx{K59}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx3{J28}
K65
NAND_4
/CPU_$5203{K65}
/Reg_Access{K74}
L84
NOR_2
L83
NAND_2
+CPU_$5203_Access{L84}
/CPU_$5203_WR{L83}
+Reg_Write{L85}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx4{J26}
K66
NAND_4
/CPU_$5204{K66}
+Reg_Read{K3}
L86
NAND_2
/Reg_Access{K74}
L88
NOR_2
+CPU_$5204_Access{L88}
L87
NAND_2
+Reg_Write{L85}
/CPU_$5204_WR{L87}
/CPU_$5204_RD{L86}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx5{J24}
K67
NAND_4
/CPU_$5205{K67}
+Reg_Read{K3}
L90
NAND_2
/Reg_Access{K74}
L91
NOR_2
+CPU_$5205_Access{L91}
L89
NAND_2
+Reg_Write{L85}
/CPU_$5205_WR{L89}
/CPU_$5205_RD{L90}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K68}
+CPU_$xxx6{J22}
K69
NAND_4
/CPU_$5206{K69}
+Reg_Read{K3}
L94
NAND_2
/Reg_Access{K74}
L92
NOR_2
+CPU_$5206_Access{L92}
L93
NAND_2
+Reg_Write{L85}
/CPU_$5206_WR{L93}
/CPU_$5206_RD{L94}
+CPU_$5xxx{K84}
+CPU_$xCxx-xFxx{J44}
K95
NAND_2
/Reg_Access{K98}
L117
NOR_2
+CPU_ExRAM_Access{L117}
+CPU_$A1{H5}
/CPU_$A2{H8}
+CPU_$A3{H9}
+CPU_A4{H11}
+CPU_A5{H13}
+CPU_A6{H15}
+CPU_A7{H17}
J4
NAND_7
+CPU_A8{H19}
+CPU_A9{H21}
+CPU_A10{H23}
+CPU_A11{H25}
+CPU_A12{H27}
+CPU_A13{H29}
+CPU_A14{H31}
J5
NAND_7
J6
NOR_2
J7
INVERT
+Reg_Access{H36}
J8
NOR_2
+CPU_$FFFA/B_Access{J8}
L21
NAND_2
+Reg_Read{K3}
/CPU_$FFFA/B_RD{L21}
/CPU_$5120_WR{L56}
/CPU_$5121_WR{L58}
/CPU_$5122_WR{L62}
/CPU_$5123_WR{L64}
/CPU_$5124_WR{L66}
/CPU_$5125_WR{L69}
/CPU_$5126_WR{L70}
/CPU_$5127_WR{L74}
AF1
NAND_8
AE1
INVERT
/CPU_$5120/1/2/3/4/5/6/7_WR{AE1}
/CPU_$5128_WR{L98}
/CPU_$5129_WR{L99}
/CPU_$512A_WR{L102}
/CPU_$512B_WR{L104}
AF2
NAND_4
AE2
INVERT
/CPU_$5128/9/A/B_WR{AE2}
Sheet: /Register Access/
File: register_access.kicad_sch
Title:
Size: USLedger
Date:
Rev:
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Id: 3/26

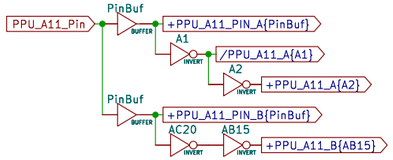
PPU Address Bus Buffering

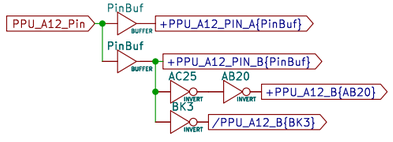


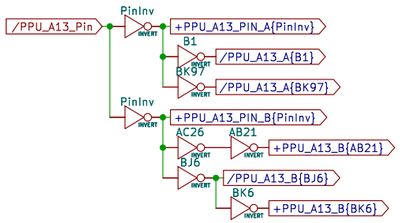


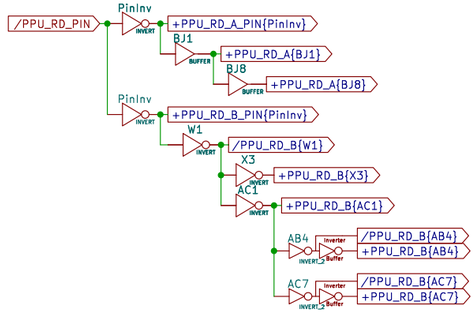


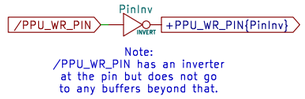


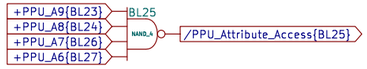






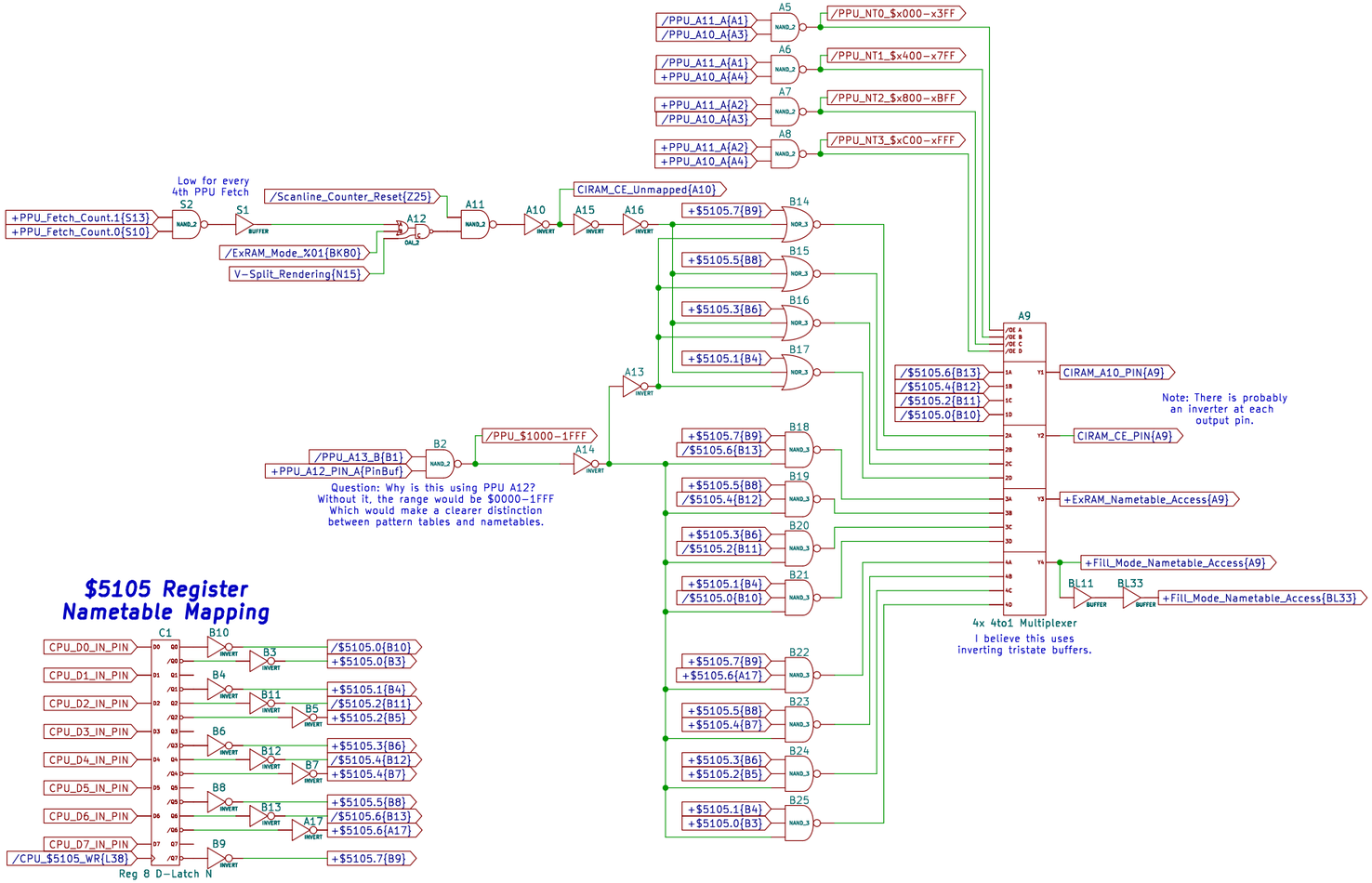






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File: ppu_address_bus.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
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$5105 Register Nametable Mapping

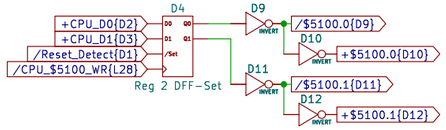


Sheet: /Nametable Mapping/
File: nametable_mapping.kicad_sch

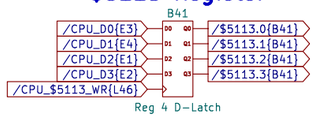
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Size: USLedger | Date: | **Rev:**
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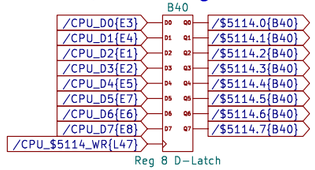
$5100 Register
PRG Mode



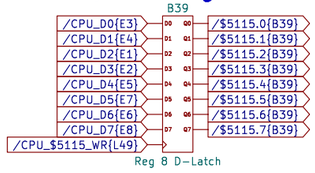
PRG Bank Registers
$5113 Register



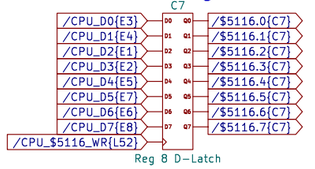
$5114 Register



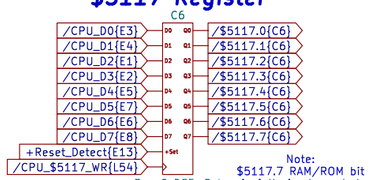
$5115 Register



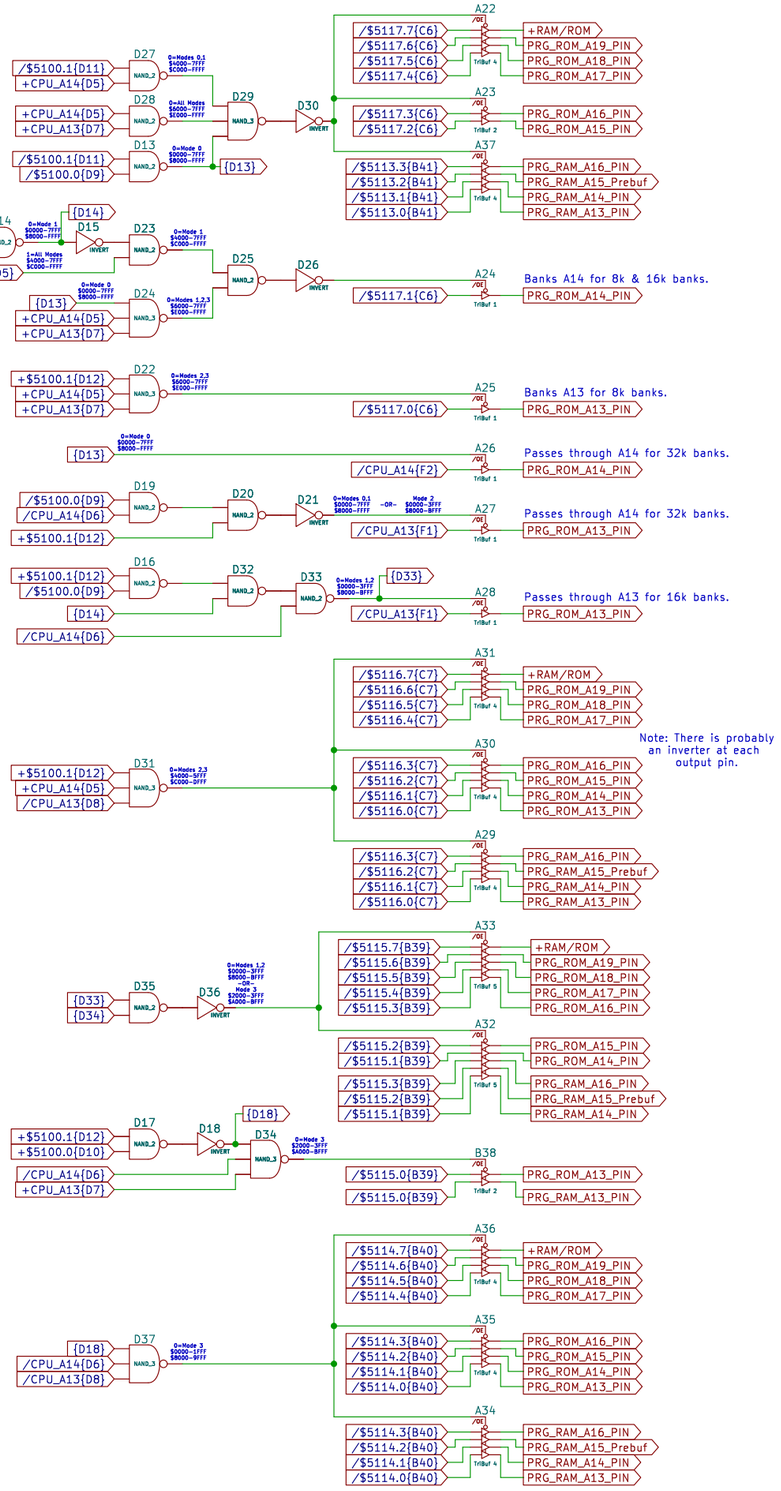
$5116 Register



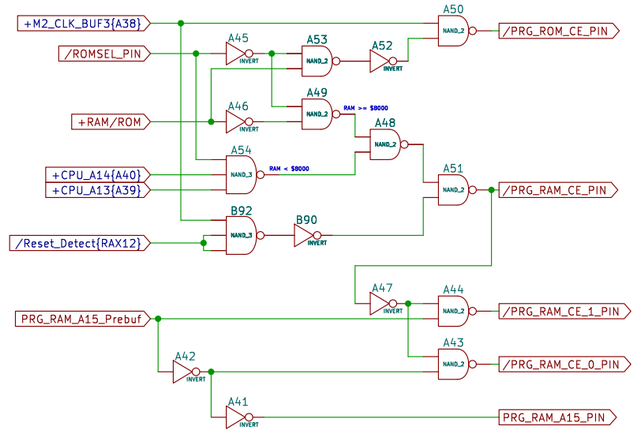
$5117 Register



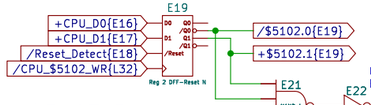
Note: $5117.7 RAM/ROM bit is fully implemented but is somehow being prevented from entering RAM mode when bench testing.



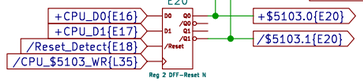
Note: There is probably an inverter at each output pin.



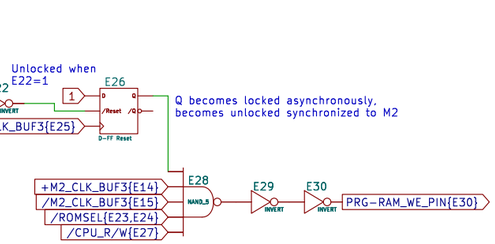
$5102 Register
PRG-RAM Write Protection



$5103 Register



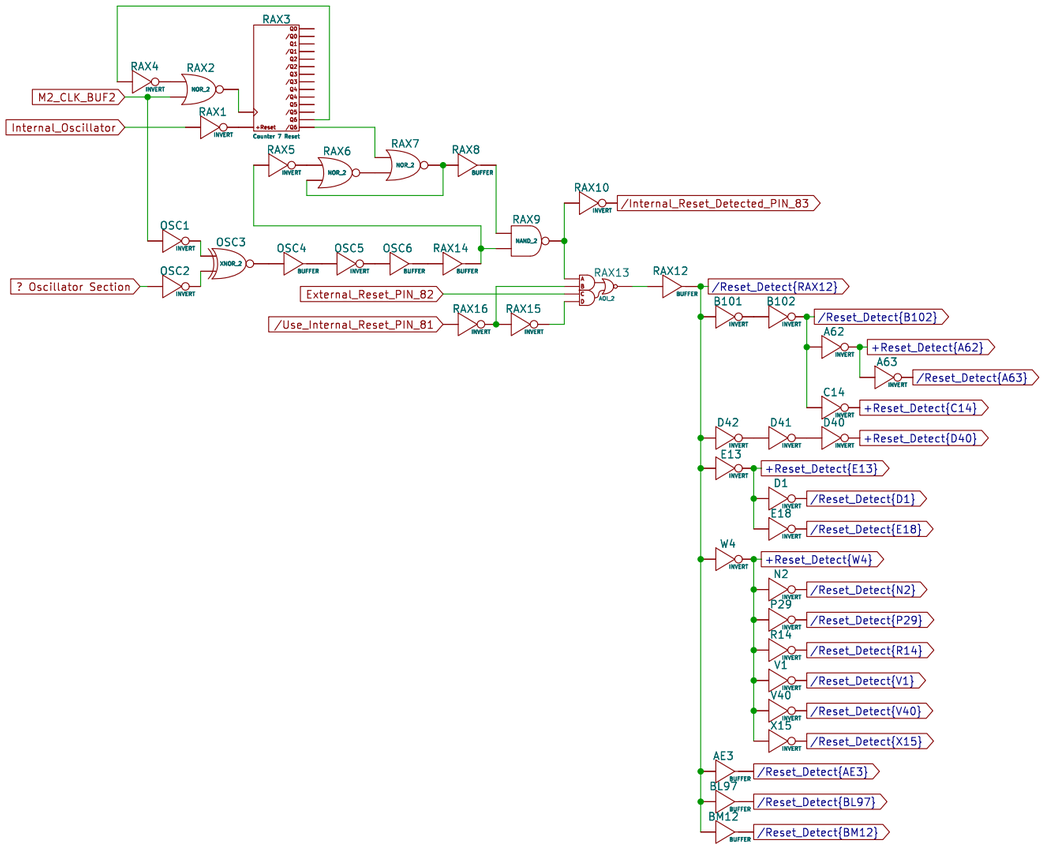
Note: /Set vs. /Reset inputs are individually used but not easily shown here with a symbol.

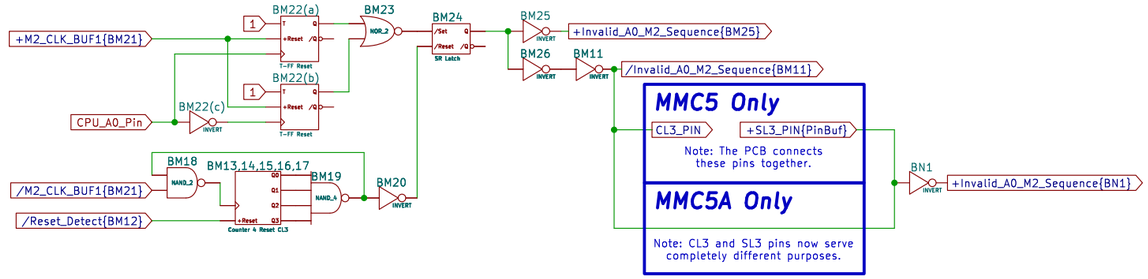


Sheet: /PRG Banking/
File: prg_banking.kicad_sch

Title:

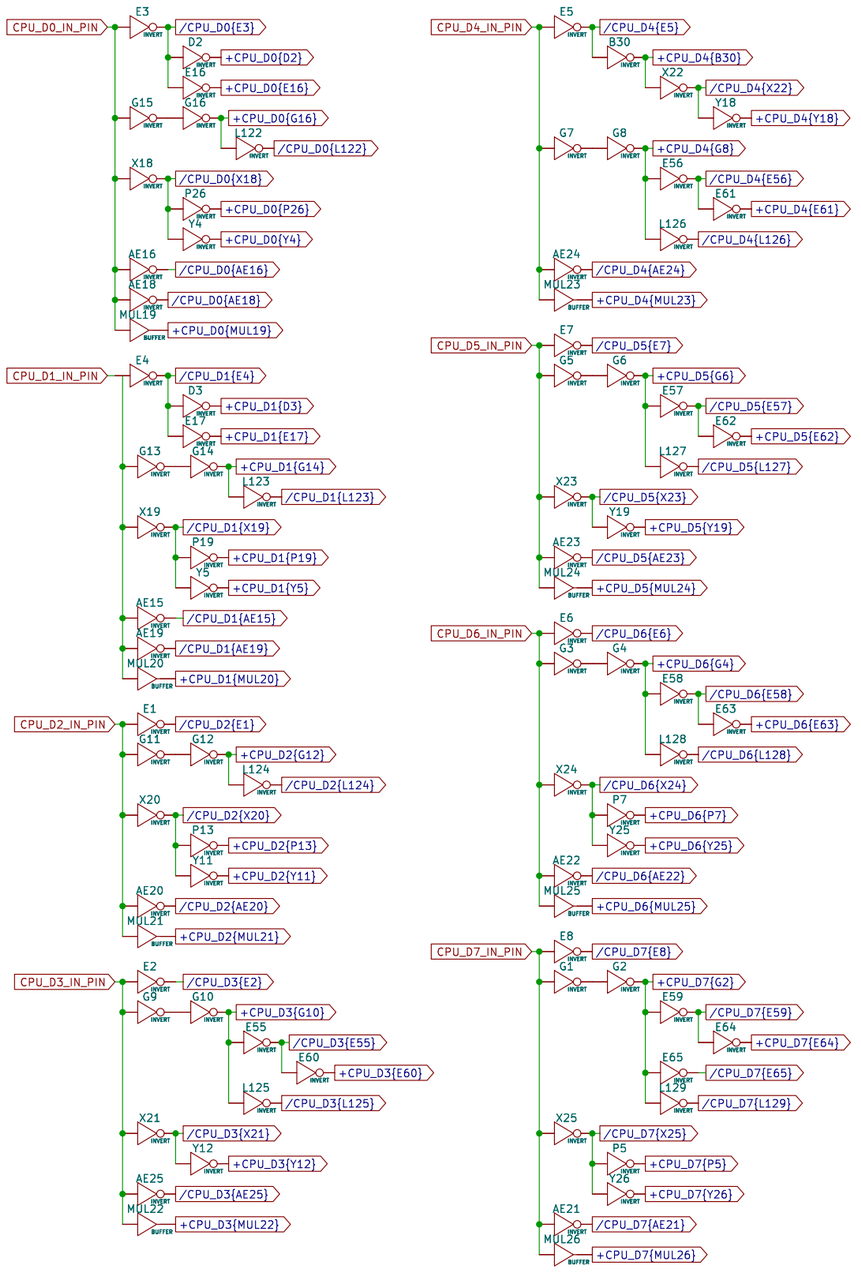
Size: USLedger | Date: | **Rev:**
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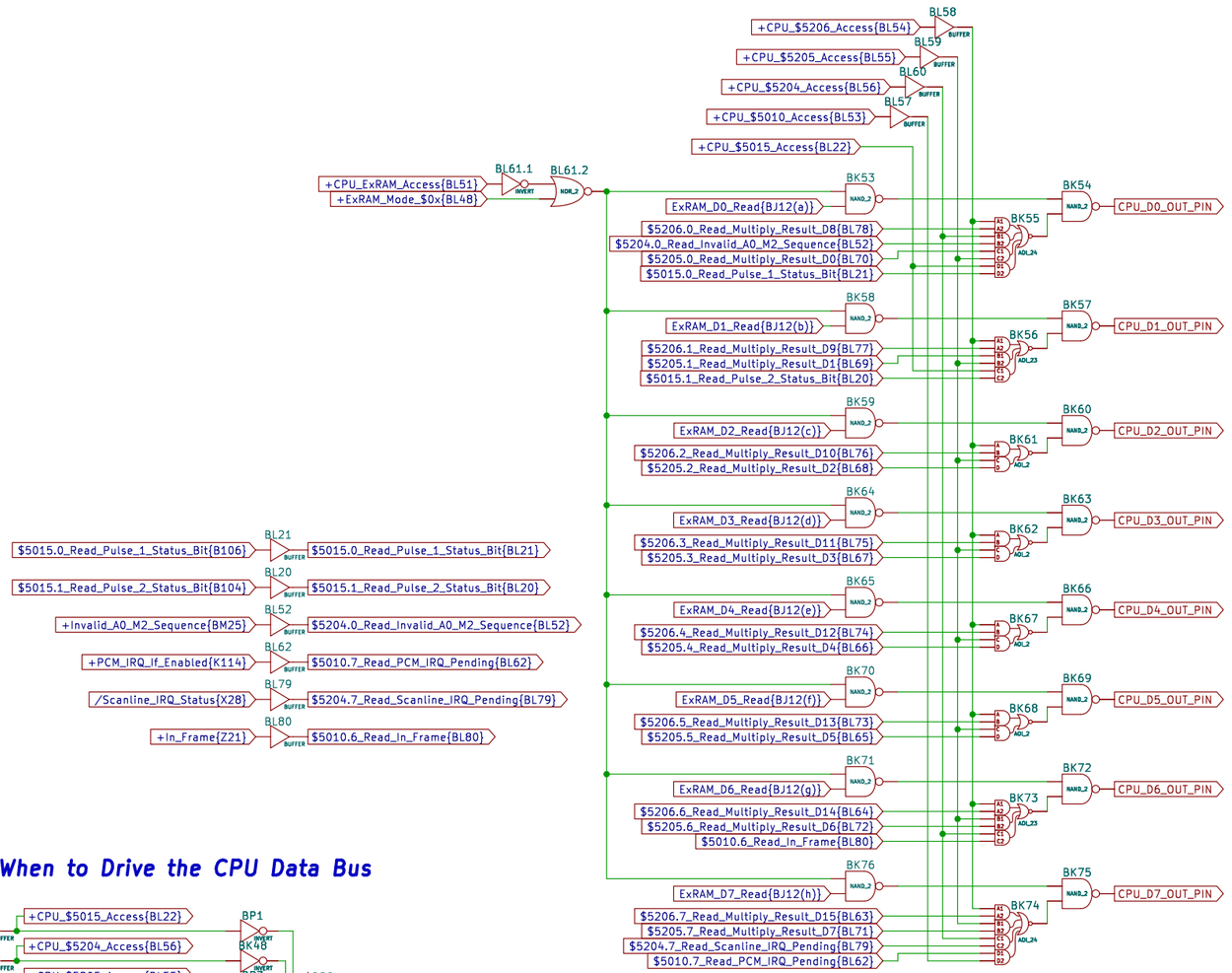


Sheet: /Reset Detection/	
File: reset_detection.kicad_sch	
Title:	
Size: USLedger	Date:
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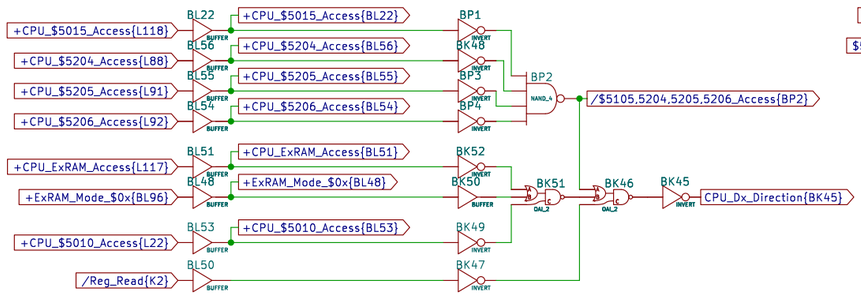
Incoming CPU Data Bus Buffering Bonanza



Multiplexing Data to be Driven onto the CPU Data Bus

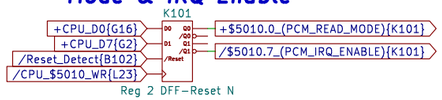


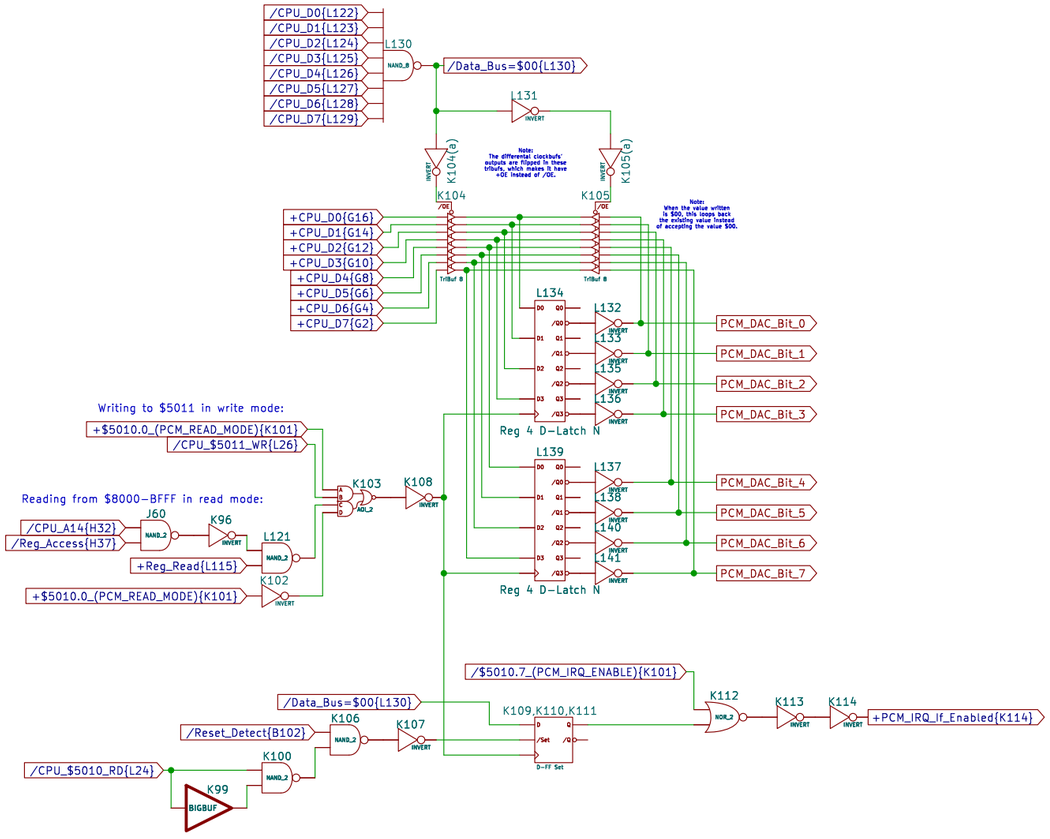
Decides When to Drive the CPU Data Bus



Sheet: /CPU Data Bus/		
File: cpu_data_bus.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
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$5010 Register
Mode & IRQ Enable





Sheet: /PCM DAC/	
File: pcm_dac.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
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Scanline Start Detection

+Use Internal Scanline Detection

+Use External Scanline Detection

+Test_Mode_PIN_92{Pininv} → AA1 (INVERT, Inverter Buffer)

/Test_Input_PIN_29{Pininv} → AA4 (INVERT)

External Scanline Start

High when the present PPU address is from $2xxx and matches the previous read.

AA5 (NOR_4) ← +PPU_A13_B{AB21}, +PPU_A12_B{AB20}, +PPU_A11_B{AB15}, +PPU_A10_B{AB14}, +PPU_A9{AB9}, +PPU_A8{AB8}, +PPU_A7{AB5}, +PPU_A6{AC10}, +PPU_A5{AC11}, +PPU_A4{AC16}, +PPU_A3{AC17}, +PPU_A2{AC22}, +PPU_A1{AC23}, +PPU_A0{AC28}

+PPU_A13_B{AB21} → AB22 (D-FF Reset) → AA21 (NAND_2) → AA20 (INVERT)

+PPU_A12_B{AB20} → AB19 (D-FF Reset) → AA18 (NOR_2)

+PPU_A11_B{AB15} → AB16 (D-FF Reset) → AA17 (XNOR_2)

+PPU_A10_B{AB14} → AB13 (D-FF Reset) → AA14 (XNOR_2)

AA16 (NAND_4) ← AA20, AA17, AA14

+PPU_A9{AB9} → AB10 (D-FF Reset) → AA13 (XNOR_2)

+PPU_A8{AB8} → AB7 (D-FF Reset) → AA12 (XNOR_2)

+PPU_A7{AB5} → AB6 (D-FF Reset) → AA9 (XNOR_2)

+PPU_A6{AC10} → AC9 (D-FF Reset) → AA8 (XNOR_2)

AA10 (NAND_4) ← AA13, AA12, AA9, AA8

+PPU_RD_B{AB4} → AA7 (INVERT)

/Exit_In_Frame{AC6} → AA6 (INVERT)

+PPU_RD_B{AC7} → AB3 (INVERT)

When not in frame, the previous PPU read address always looks like $0000. Since A13, A12 are looking at range $2xxx instead of matches, $0xxx won't match $2xxx. Therefore, scanlines don't get detected when using PPUDATA to read from $0000 during v-blank.

+PPU_A5{AC11} → AC12 (D-FF Reset) → AA11 (XNOR_2)

+PPU_A4{AC16} → AC15 (D-FF Reset) → AB11 (XNOR_2)

+PPU_A3{AC17} → AC18 (D-FF Reset) → AB12 (XNOR_2)

AA15 (NAND_4) ← AA11, AB11, AB12

+PPU_A2{AC22} → AC21 (D-FF Reset) → AB17 (XNOR_2)

+PPU_A1{AC23} → AC24 (D-FF Reset) → AB18 (XNOR_2)

+PPU_A0{AC28} → AC27 (D-FF Reset) → AA22 (XNOR_2)

AA19 (NAND_2) ← AB17, AB18, AA22

AA5 → AB2 (SR Reg 3) → AB1 (NAND_4)

/PPU_RD_B{W1} → AB2

AB2(a) (INVERT)

Low when the 3 most recent PPU reads were from $2xxx and all matched.

AA1, AA4, AB1 → AA2 (AOI_2) → AA3 (INVERT) → /Scanline_Start_Detected{AA3}

/Scanline_Start_Detected{AA3} → R2 (BUFFER) → /Scanline_Start_Detected{R2}

M2_CLK_BUF1 → AC2 (Counter 2 Reset)

+PPU_RD_B{AC1} → AC2

AC2 → AC6 (NAND_2) → /Exit_In_Frame{AC6}

Scanline End Detection

Increment_Scanline_Fetch_Count{S6}, +PPU_Cycle_336{R4} → R6 (NAND_2) → R7 (INVERT) → R8 (INVERT) → Scanline_End_Detected_Internal{R8}

Scanline_End_Detected_Internal{R8}, /Test_Input_PIN_30{Pininv} → R1 (AOI_3) → P3 (INVERT) → Scanline_End_Detected{P3}

P3 → T5 (BUFFER) → Scanline_End_Detected{T5}

/Test_Input_PIN_30{Pininv} → P4 (INVERT)

External Scanline End

$5204 Register Scanline IRQ Enable

+CPU_D7{Y26}, /Reset_Detect{X15}, /CPU_$5204_WR{L87} → X12 (D-FF Reset) → X13 (NAND_2) → X14 (BUFFER) → +Scanline_IRQ_Pin{X14}

$5203 Register IRQ Scanline Compare Value

+CPU_D0{Y4}, +CPU_D1{Y5}, +CPU_D2{Y11}, +CPU_D3{Y12}, +CPU_D4{Y18}, +CPU_D5{Y19}, +CPU_D6{Y25}, +CPU_D7{Y26} → Y1 (BUFFER, Reg 8 D-Latch)

/CPU_$5203_WR{L83} → Y1

Y2,7,9,14,16,21,23,28

Y1 → Y3, Y6, Y10, Y13, Y17, Y20, Y24, Y27 (XNOR_2)

Y3, Y6, Y8 → Y8 (NAND_4)

Y10, Y13, Y17, Y20, Y22 → Y22 (NAND_4)

Y8, Y22, Y24, Y27 → Y15 (NOR_2)

Hi when $5203 matches the present V-Split scanline.

D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

IRQ Scanline Counter

Z2-Z18 (Counter 8 Reset)

/Scanline_Start_Detected{AA3}, +In_Frame{Z21} → Z1 (NAND_2) → Z19 (BUFFER) → Z2-Z18

/In_Frame{Z23} → Z19

Z22 (NAND_4) ← Z2-Z18

Z22 → /Scanline_240_and_up{Z22}

$2001 Register Write Snooping

+CPU_D3{Y12}, +CPU_D4{Y18}, /CPU_$2001_WR{L107} → X8 (Reg 2 DFF-No-Reset N) → X9 (NAND_2)

Hi = PPU Rendering Enabled

$2002 Register Read Snooping

Latches $2002.7 value read by CPU for 1 CPU cycle.

/CPU_$FFFA/B_RD{L21}, /CPU_$4014_WR{L116}, /Exit_In_Frame{AC6} → X6 (NAND_4) → X7 (INVERT) → Z23 (D-FF Reset)

/CPU_$2002_RD{L110} → X16 (INVERT) → X4 (NAND_2)

+CPU_D7{Y26} → X4

M2_CLK_BUF1 → X5 (D-FF No Reset)

X4 → X5

/Scanline_Start_Detected{AA3} → Z23

Z23 → /In_Frame{Z23}, Z21 (INVERT) → +In_Frame{Z21}

/In_Frame{Z23} → Z24 (NAND_2) → Z25 (INVERT) → +Rendering_In_Frame{Z25}

Z25 → V39 (INVERT) → /Rendering_In_Frame{V39}

Z25 → R24 (INVERT) → /Rendering_In_Frame{R24}

Z25 → N16 (INVERT) → N9 (INVERT) → +Rendering_In_Frame{N9}

M3 (BIGBUF) → N38 (NAND_2) → N39 (INVERT) → M2 (NAND_2)

/CPU_$5204_RD{L86} → M3

+New_Scanline_Detected{AA3 via Z23}, /In_Frame{Z23} → Z20 (NAND_2) → X17 (NAND_2) → X26 (INVERT) → Y31 (D-FF Reset)

/Scanline_Start_Detected{AA3} → Z20

+IRQ → Y31

Y31 → Y29 (NOR_2) → Y30 (INVERT) → /CPU_$5204_RD{L86}

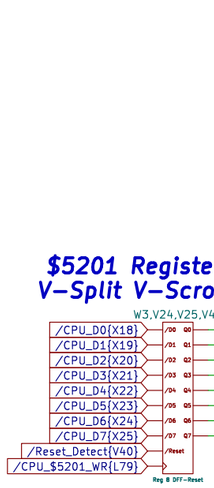
+IRQ → Y32 (INVERT) → X27 (D-FF No Reset) → X28 (INVERT) → /Scanline_IRQ_Status{X28}

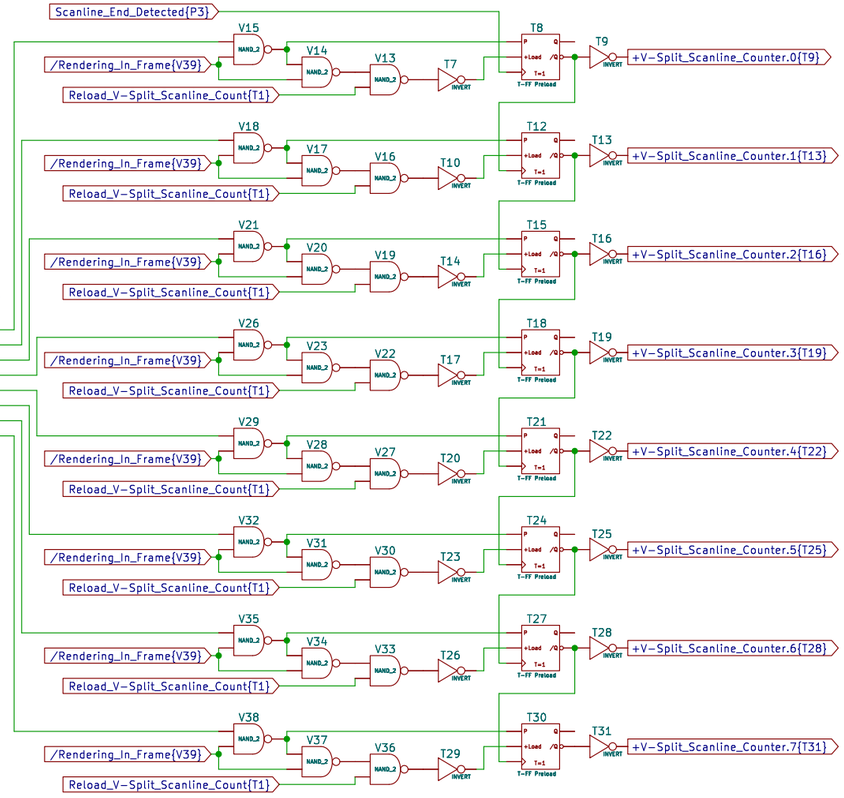
/IRQ

Sheet: /Scanline Detection/	
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Title:	
Size: B	Date:
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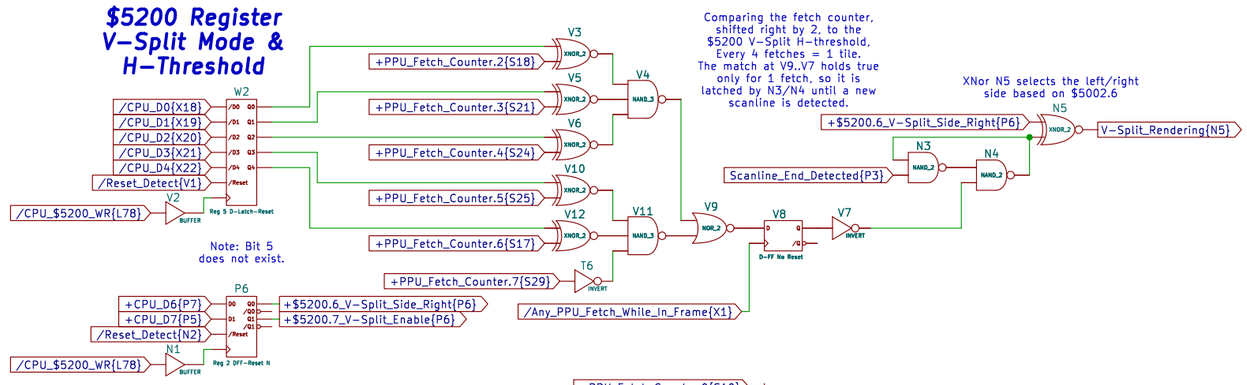
V−Split Scanline Counter

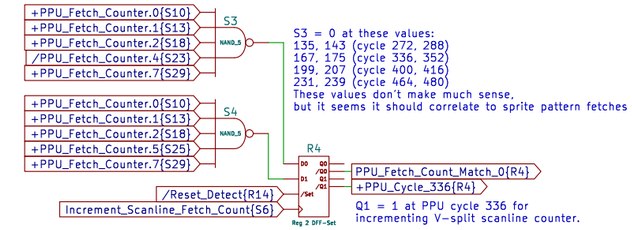
$5201 Register V−Split V−Scroll

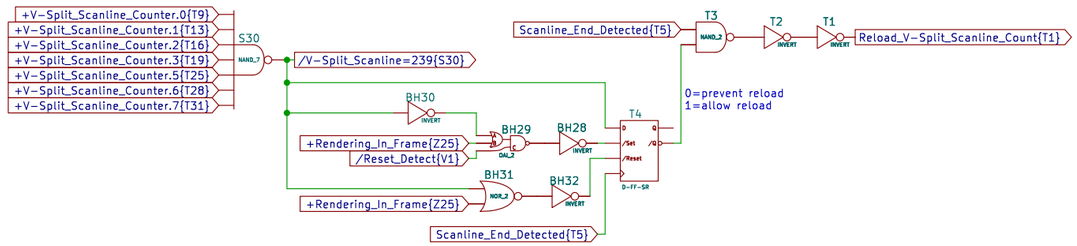




$5200 Register V−Split Mode & H−Threshold





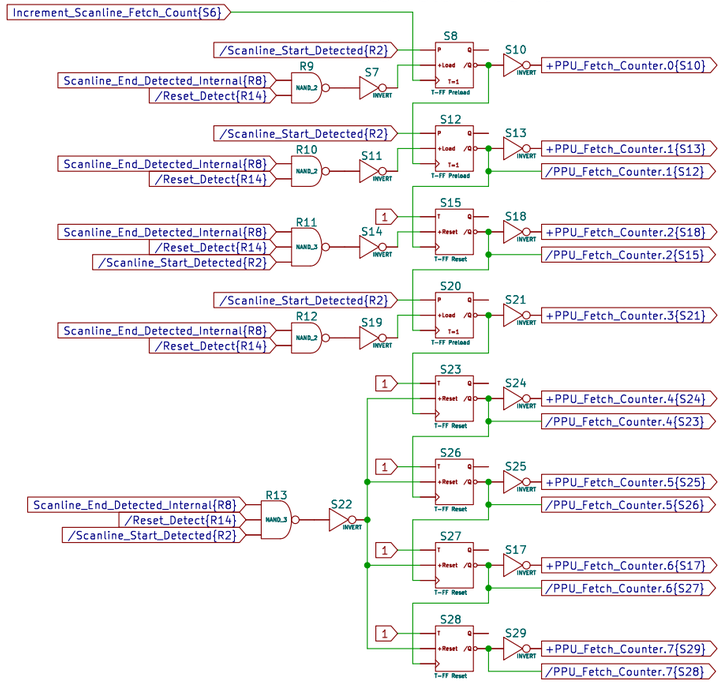


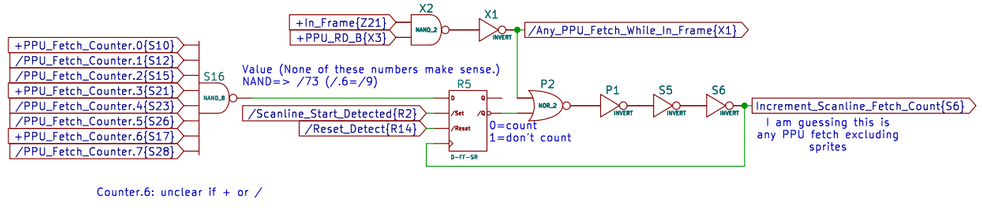
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File: v_split.kicad_sch

Title:

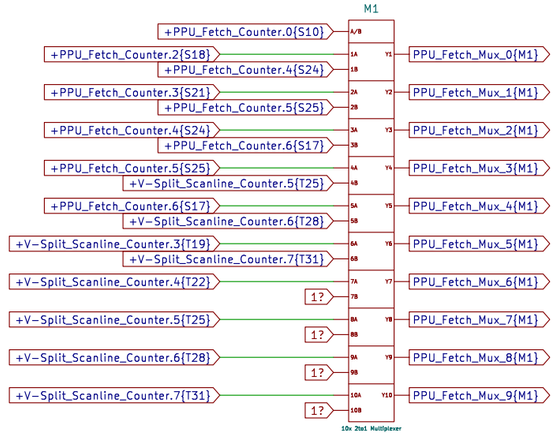
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PPU Fetch Counter





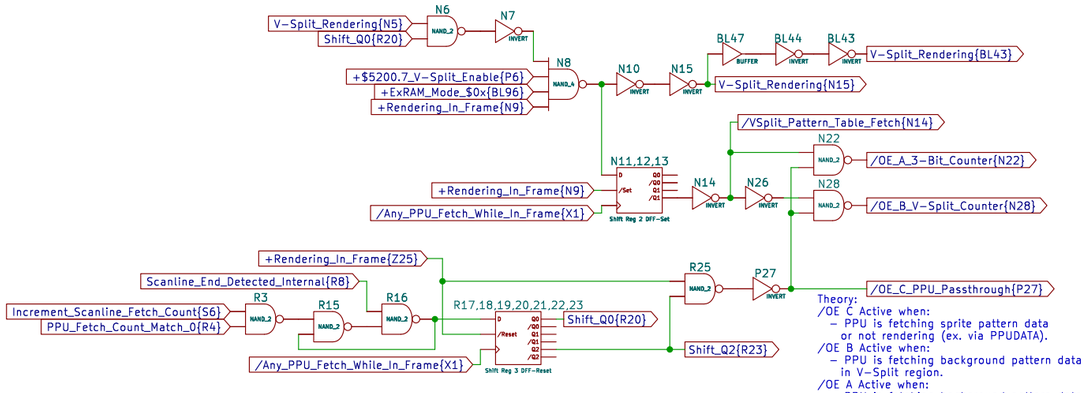
ExRAM Attribute Fetch Mux

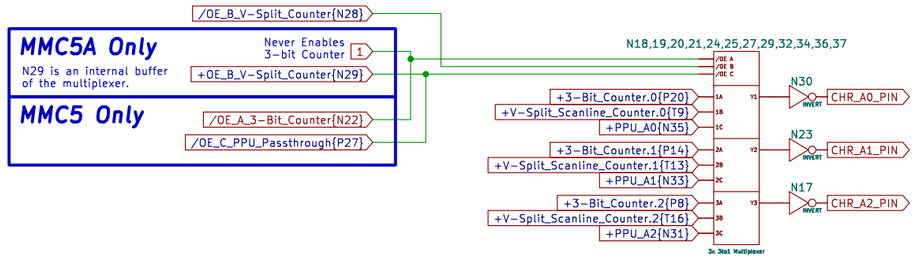


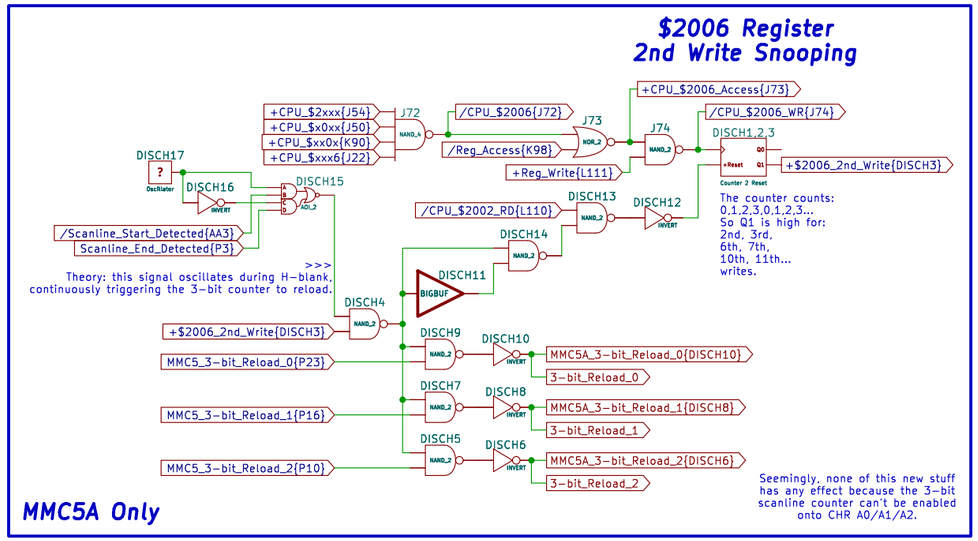
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File: ppu_fetch_counting.kicad_sch

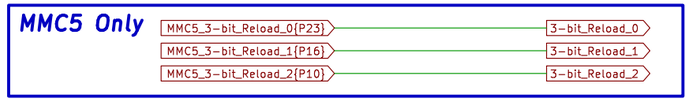
Title:

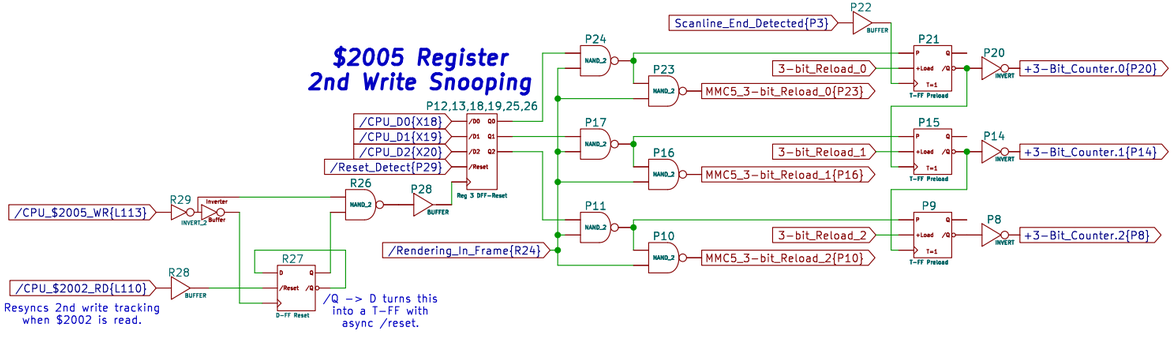
Size: USLedger	Date:	**Rev:**
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Sheet: /CHR A0,A1,A2/		
File: chr_a0_a1_a2.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
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$5104 Register
ExRAM Mode

CPU_D0_IN_PIN
CPU_D1_IN_PIN
/Reset_Detect{BL97}
/CPU_$5104_WR{L37}
BK82
Reg 2 DFF-Set
BK79
BK87
+ExRAM_Mode_%10{BK87}
BK80
BL49
/ExRAM_Mode_%01{BK80}
/ExRAM_Mode_%01{BL49}
BK81
BL96
/ExRAM_Mode_%00{BK81}
Appears to go nowhere.
+ExRAM_Mode_$0x{BL96}
Tracks whether $5104 has been written since reset detection
/Reset_Detect{RAX12}
M2_CLK_BUF1
+Invalid_A0_M2_Sequence{BN1}
BL89
BL90
BL91
SR Latch
Syncs SL3 to M2
BL98
D-FF Reset
BL94
BL92
BL93
BN2
BIGBUF
There appears to be a selectable delay jumper, connected at BM6.
BM10 BM9 BM8 BM7 BM6 BM5 BM4 BM3 BM2 BM1
BUFFER
+ExRAM_Mode_$0x{BL96}
BL86
BL88
BL85
BL84
BL83
PPU_RW_Mode_%0x_Access{BL83}
Not sure about polarity; this may be Mode %1x Access, and active low or high.
+PPU_RD_A_PIN{PinInv}
+PPU_WR_PIN{PinInv}
BL82
BL87
BL81
BIGBUF

+CPU_ExRAM_Access{L117}
+Reg_Write{BK88}
BK85
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BK86
+ExRAM_Mode_%10{BK87}
Writing to ExRAM via $5C00-5FFF in mode %10
Writing to an ExRAM nametable via PPU via $2000
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL96}
+PPU_WR_PIN{PinInv}
BK83
BK84
+ExRAM_WR{BK84}
This is the write signal for the ExRAM itself.

BN4
BIGBUF
BL105
BL106
BL109
/Q{BL109}
+ExRAM_Mode_$0x{BL96}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BL108
BL99
/Q{BL99}
/PPU_A13_A{BK97}
BL102
BL101
BL100
/Reset_Detect{RAX12}
BN3
BIGBUF
BL107
+PPU_RD_A_PIN{PinInv}

/PPU_A13_A{BK97}
+ExRAM_Mode_$0x{BL96}
BK94
BK95
+Drive_ExRAM_Ax_From_PPU_Ax{BK95}
V-Split_Rendering{N15}
BK92
BK93
BK91
+Drive_ExRAM_Ax_For_Substituted_Fetch{BK91}
BK96
BJ51
+Drive_ExRAM_Ax_From_CPU_Ax{BJ51}
TRUE when:
ExRAM is mode $1x, which gives CPU access but no PPU access,
OR PPU is accessing NT or AT
That doesn't make sense because what if you have a nametable assigned to ExRAM in $5105? It needs ExRAM Ax coming from PPU.
Signal A9 (+ExRAM Nametable Access) is VERY closeby but no obvious connection
Consider:
ExRAM NEVER contains pattern tables and that should be related to how PPU /A13 is being used here.
Question:
Why should we care about PPU /A13 AT ALL when the CPU has control of ExRAM address bus?
Consider:
ExRAM can still be used as a nametable when ExRAM mode is set to CPU mode.

BJ50 BJ49 BJ48 BJ47 BJ46 BJ45 BJ44 BJ43 BJ42 BJ41
NAND Mux 3
PPU_Fetch_Mux_0{M1} … PPU_Fetch_Mux_9{M1}
PPU_A0_Pin … PPU_A9_Pin
ExRAM_A0{BJ50} … ExRAM_A9{BJ41}

BJ40
Reg 10 D-Latch
CPU_A0_Pin … CPU_A9_Pin

/Q{BL109}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
+ExRAM_Mode_$0x{BL96}
BK89
MMC5 Only
+PPU_RD_A_PIN{PinInv}
MMC5A Only
/Q{BL99}
BK90

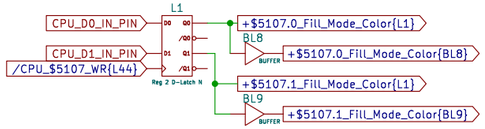
BK78
BK77
CPU_D0_IN_PIN … CPU_D7_IN_PIN
CPU_D0_In_Latched{BK77} … CPU_D7_In_Latched{BK77}
Reg 8 D-Latch
Reg 8 D-Latch

/Scanline_Counter_Reset{Z25}
+ExRAM_Mode_$0x{BL96}
BL95
BL104
BL103
BJ39b
BJ39 BJ38 BJ37 BJ36 BJ35 BJ34 BJ33 BJ32
NAND Mux 2
CPU_D0_In_Latched{BK77} … CPU_D7_In_Latched{BK77}
PPU_D0_In{BigBuf} … PPU_D7_In{BigBuf}
ExRAM_D0_Write{BJ39} … ExRAM_D7_Write{BJ32}

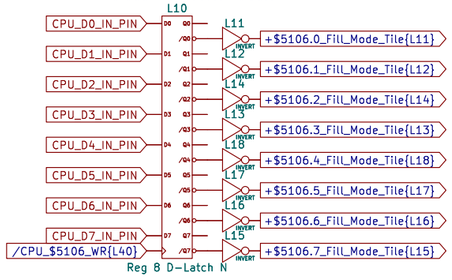
ExRAM_D0_Read … ExRAM_D7_Read
BJ12(a) … BJ12(h)
BUFFER
ExRAM_D0_Read{BJ12(a)} … ExRAM_D7_Read{BJ12(h)}
BJ11
/ExRAM_Read_D0_Latch{BJ11} … /ExRAM_Read_D5_Latch{BJ11}
+ExRAM_Read_D6_Latch{BJ11}
+ExRAM_Read_D7_Latch{BJ11}
+PPU_RD_A{BJ8}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.0{S10}
BJ9
BJ10
BJ7
Reg 8 D-Latch N

Sheet: /ExRAM Mode/
File: exram_mode.kicad_sch
Title:
Size: USLedger | Date:
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Rev:
Id: 14/26

$5107 Register
Fill-Mode Color



$5106 Register
Fill-Mode Tile



Sheet: /Fill Mode/
File: fill_mode.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 15/26

5-Bit to 7-Bit Pulse Length Lookup Table

See spreadsheet for decoded values:
"pulse length lookup table.xlsx"

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D46 (NAND_5) → D45 (INVERT)

+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D48 (NAND_4) → D47 (INVERT)

D45, D47 → C16 (NOR_2) → C15 (INVERT) → Pulse_LUT_D6{C15}

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
→ D50 (NAND_5) → D49 (INVERT)

+CPU_D7{E64}
+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D3{E55}
→ D52 (NAND_4) → D51 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
→ D54 (NAND_5) → D53 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D56 (NAND_5) → D55 (INVERT)

D49, D51, D53, D55 → C18 (NOR_4) → C17 (INVERT) → Pulse_LUT_D5{C17}

+CPU_D7{E64}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E61}
/CPU_D3{E55}
→ D57 (NAND_5) → D58 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D59 (NAND_4) → D60 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D61 (NAND_5) → D62 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D63 (NAND_5) → D64 (INVERT)

D58, D60, D62, D64 → C20 (NOR_5) → C19 (INVERT) → Pulse_LUT_D4{C19}

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D65 (NAND_5) → D66 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D4{E61}
/CPU_D3{E55}
→ D67 (NAND_5) → D68 (INVERT)

/CPU_D7{E59}
+CPU_D4{E61}
/CPU_D3{E55}
→ D69 (NAND_3) → D70 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D71 (NAND_5) → D72 (INVERT)

+CPU_D7{E64}
+CPU_D3{E60}
→ D73 (NAND_2) → D74 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D4{E56}
→ D75 (NAND_4) → D76 (INVERT)

D68, D70, D72, D74, D76 → C22 (NOR_5) → C21 (INVERT) → Pulse_LUT_D3{C21}

+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D77 (NAND_4) → D78 (INVERT)

/CPU_D7{E59}
+CPU_D6{E62}
/CPU_D4{E56}
/CPU_D3{E55}
→ D79 (NAND_4) → D80 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D81 (NAND_4) → D82 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
→ D83 (NAND_3) → D84 (INVERT)

+CPU_D6{E63}
+CPU_D3{E60}
→ D85 (NAND_2) → D86 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
→ D87 (NAND_4) → D88 (INVERT)

D78, D80, D82, D84, D86, D88 → C24 (NOR_6) → C23 (INVERT) → Pulse_LUT_D2{C23}

+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D89 (NAND_5) → D90 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D91 (NAND_5) → D92 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E56}
/CPU_D3{E55}
→ D93 (NAND_5) → D94 (INVERT)

+CPU_D5{E62}
+CPU_D3{E60}
→ D95 (NAND_2) → D96 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D97 (NAND_5) → D98 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D99 (NAND_5) → D100 (INVERT)

D90, D92, D94, D96, D98, D100 → C26 (NOR_6) → C25 (INVERT) → Pulse_LUT_D1{C25}

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D3{E55}
→ D101 (NAND_4) → D102 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
→ D103 (NAND_4) → D104 (INVERT)

+CPU_D4{E61}
+CPU_D3{E60}
→ D105 (NAND_2) → D106 (INVERT)

D102, D104, D106 → C28 (NOR_3) → C27 (INVERT) → Pulse_LUT_D0{C27}

$5003 Register
Pulse 1 Length Counter

+$5000.5_Length_Halt{G85} → B44 (INVERT) → B43 (NAND_2) → B42 (NOR_2)
Pulse_1_L_Restart{C31(b)} → B43

Pulse_Length_Clock_SL3(Reset_Aware){A75}
241.08Hz

Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
→ C11 (P0 P1 P2 P3, Q0 Q1 Q2 Q3, Ripple Out, /Load Counter = Preload)

Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
/CPU_$5003_WR{K16}
→ C12 (P0 P1 P2 P3, Q0 Q1 Q2 Q3, Ripple Out, /Load Counter = Preload) → Pulse_1_L_Rollover{C12}

These counters count "down"
the preloaded period value.

This means either the outputs
or preload inputs are inverted.

$5007 Register
Pulse 2 Length Counter

+$5004.5_Length_Halt{E68} → C8 (INVERT) → C9 (NAND_2) → C10 (NOR_2)
Pulse_2_L_Restart{C31(a)} → C9

Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
→ D38 (P0 P1 P2 P3, Q0 Q1 Q2 Q3, Ripple Out, /Load Counter = Preload)

Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
/CPU_$5007_WR{K80}
→ D39 (P0 P1 P2 P3, Q0 Q1 Q2 Q3, Ripple Out, /Load Counter = Preload) → Pulse_2_L_Rollover{D39}

Pulse Channel Clock Source

1.789773 MHz
+M2_CLK_BUF2{B109}
→ A56 (NOR_2)

/Reset_Detect{A63}
1
→ A55 (T, +Reset, Q, /Q) T-FF Reset

A56, A55 → A57 (NOR_2)
894.886 kHz
<25% duty, depends on M2 Duty

A55
894.886 kHz
50% duty

A57 → A60 (INVERT) → A61 (INVERT) → Pulse_Timer_Clock_Fast{A61}
894.886 kHz
A57 → A58 (INVERT) → A59 (INVERT) → Pulse_Timer_Clock_Fast{A59}

A55 → A66 (Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7, +Reset, Q7) Counter 8 Reset
A66 → A66.2 (INVERT)
447.443 kHz
50% duty

The reset inputs of ALL these
counters seem to be /Reset.
Hard to see why this acts different
than T-FF counters elsewhere.

/Reset_Detect{A63} → A66 +Reset
3.49565 kHz
50% duty

A65 resets
Q7 of A66.

A66.3 (INVERT)
A66.6 (INVERT)
A66.7 (INVERT)
A68 (INVERT)
A66.2, A66.3, A66.6, A66.7, A68 → A66.4 (NAND_5)
/Value
3712

A67 (Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7, +Reset, Q4) Counter 5 Reset

Important:
A63 resets
Q4 of A67.

/Reset_Detect{A63} → A64 (NAND_2) → A65 (INVERT) → A67 +Reset

+Reset_Detect{A62} → A66.5 (NOR_3)
A64, A66.5 → always 0 when reset detected,
without reset, only 0 when value=3712

A66.2 in this logic can only
create a phase shift and
not affect the period
since it doesn't cause A66
to reset.

241.08 Hz
Duty seems thin

A66.4 → B108 (INVERT) → A70 (NAND_2) → A71 (INVERT) → A72 (INVERT)
241.08 Hz or
Pin 98 Input

Pulse_Timer_Clock_SL3{A72}
E51 (INVERT) → /Pulse_Timer_Clock_SL3{E51}
H87 (INVERT) → /Pulse_Timer_Clock_SL3{H87}

All of these outputs are
241.08 Hz
-or-
Pin 98 Input

A72 → A73 (NAND_2) → A74 (INVERT) → A75 (INVERT) → Pulse_Length_Clock_SL3{Reset_Aware}{A75}

/Q4 is high for the first
240Hz period after reset
has stopped being detected.

A83 (INVERT) → B107 (INVERT)

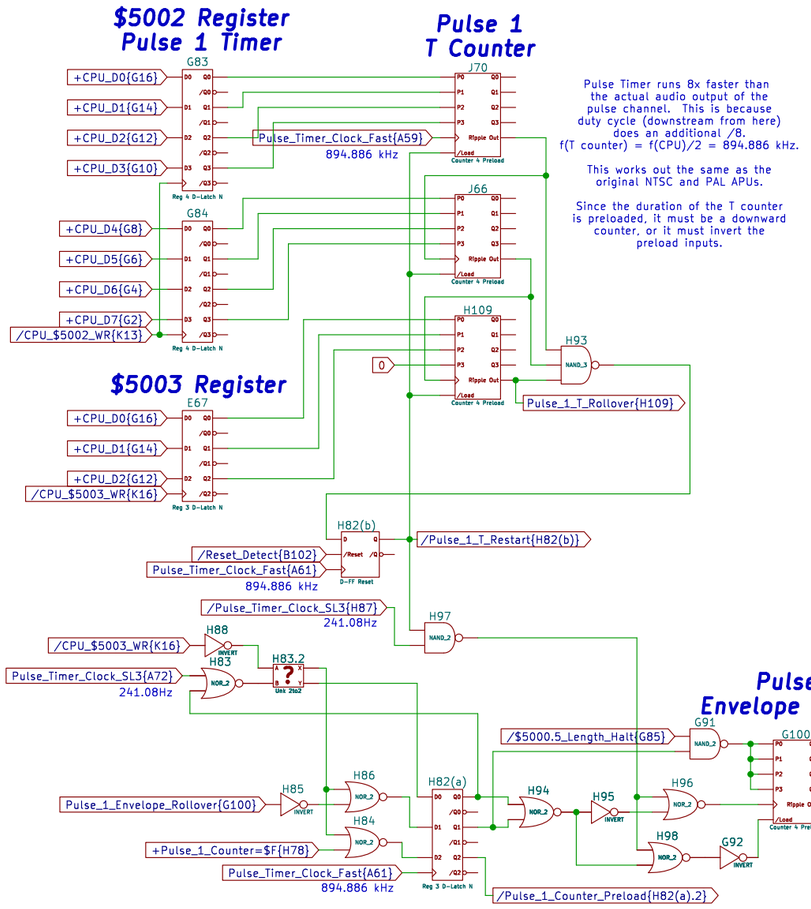
A69 (INVERT)
A79 (INVERT)
A81 (INVERT)
A69, A79, A81 → A77 (A, B, C, D, ADJ_3)
A77 → A82 (INVERT)
0=Reset Held By A67.Q4 or Pin 30

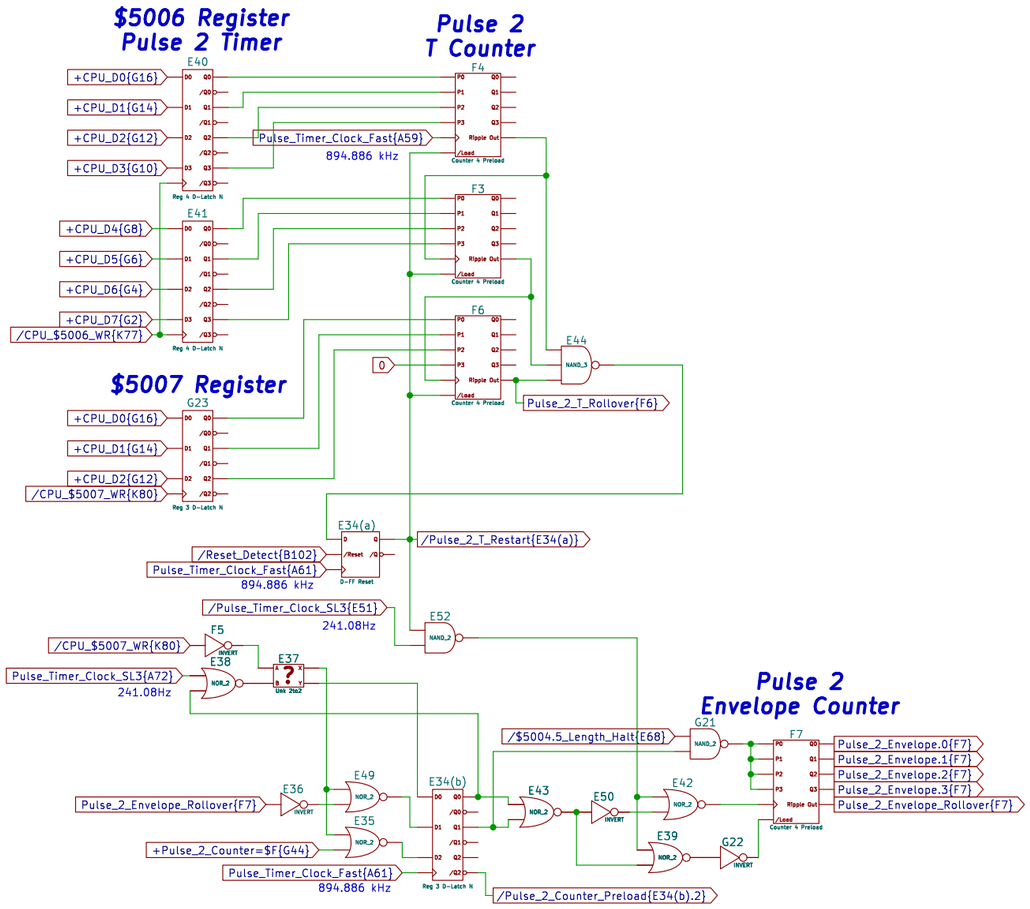
+Test_Mode_PIN_92{PinInv}
/Test_Input_PIN_30{PinInv}
→ A80 (INVERT) → A69, A79, A81

MMC5 Only
+SL3_PIN{PinBuf}

MMC5A Only
/Invalid_A0_M2_Sequence{BM11}

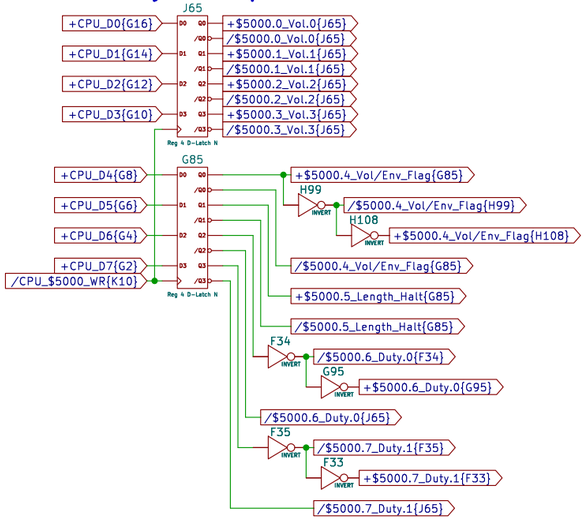
Sheet: /Pulse Length Lookup Table/	
File: pulse_length_lookup_table.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 16/26



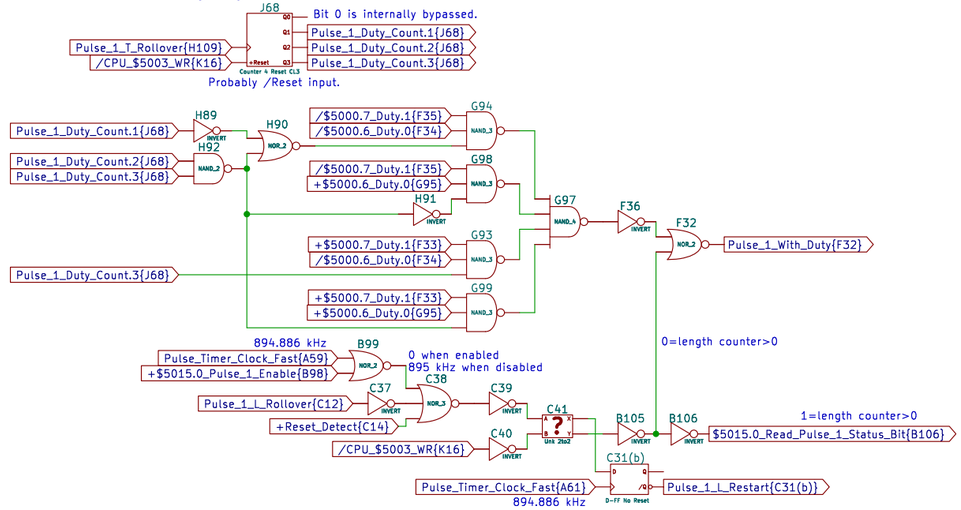


Sheet: /Pulse Timer/	
File: pulse_timer.kicad_sch	
Title:	
Size: USLedger | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 17/26

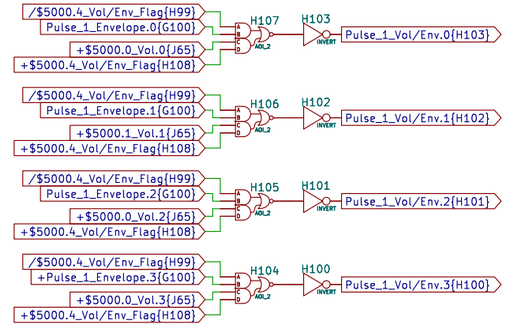
$5000 Register
Pulse 1 Duty Envelope Volume



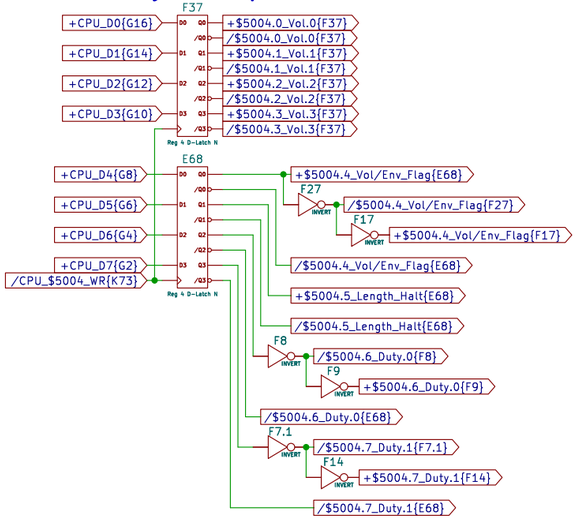
Pulse 1
Duty Cycle Counter



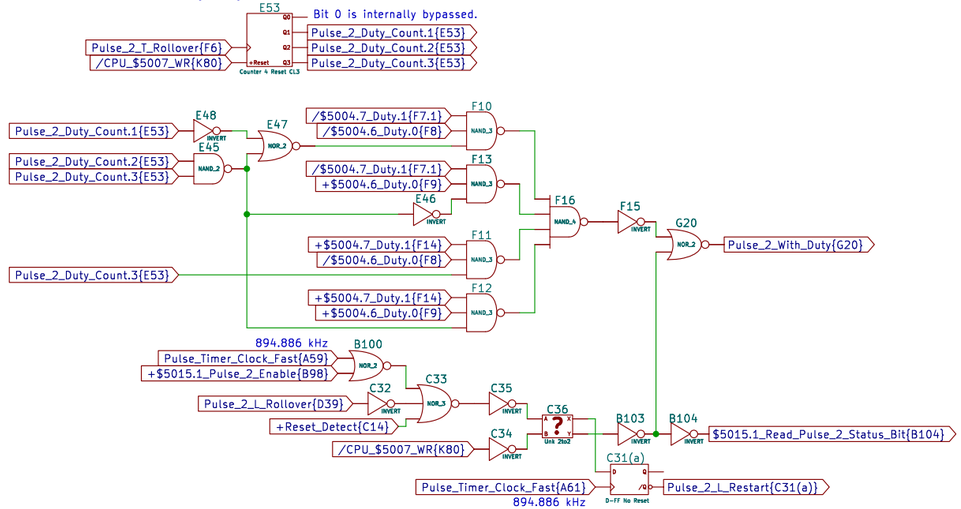
Pulse 1
Envelope/Volume Multiplexing



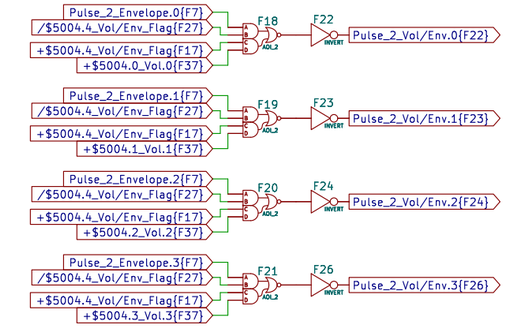
$5004 Register
Pulse 2 Duty Envelope Volume



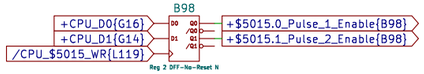
Pulse 2
Duty Cycle Counter



Pulse 2
Envelope/Volume Multiplexing



$5015 Write Register
Pulse 1&2 Status

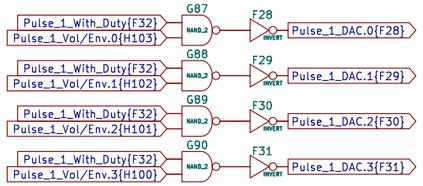


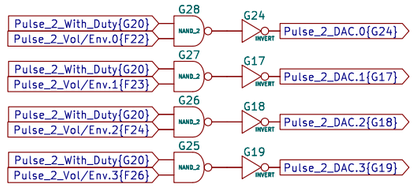
Sheet: /Pulse Duty Envelope Volume/
File: pulse_duty_envelope_volume.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 18/26

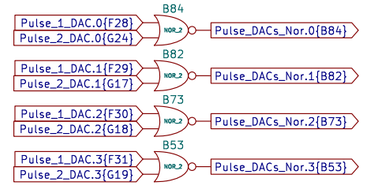
Pulse 1&2
Individual 4-bit DAC Outputs

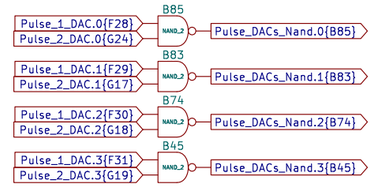




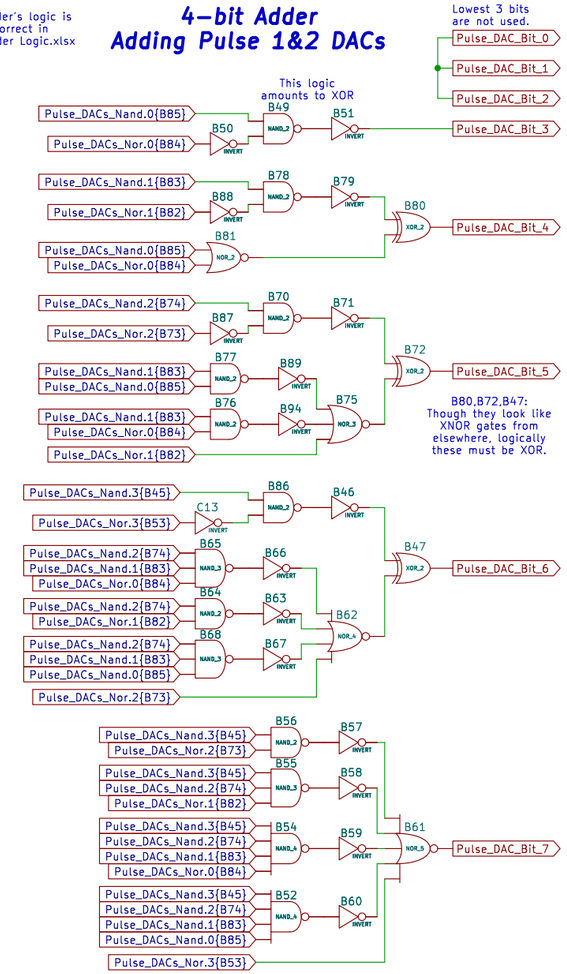
Components for
4-bit Adder

Note: The adder's logic is verified correct in Pulse DAC Adder Logic.xlsx





4-bit Adder
Adding Pulse 1&2 DACs



Sheet: /Pulse DAC/	
File: Pulse DAC.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 19/26

PPU_D0_IN_PIN
PPU_D1_IN_PIN
PPU_D2_IN_PIN
PPU_D3_IN_PIN
PPU_D4_IN_PIN
PPU_D5_IN_PIN
PPU_D6_IN_PIN
PPU_D7_IN_PIN
{BigBuf}
BIGBUF
PPU_D0_In{BigBuf}
PPU_D1_In{BigBuf}
PPU_D2_In{BigBuf}
PPU_D3_In{BigBuf}
PPU_D4_In{BigBuf}
PPU_D5_In{BigBuf}
PPU_D6_In{BigBuf}
PPU_D7_In{BigBuf}
V-Split_Rendering{BL43}
/Scanline_Counter_Reset{Z25}
/ExRAM_Mode_%01{BL49}
/Extended Attributes Enabled
+Fill_Mode_Nametable_Access{BL33}
/PPU_Attribute_Access{BL25}
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL48}
BL10
BL34
BL40
BL35
BL29
BL30
BK39
BK40
BK42
+$5107.1_Fill_Mode_Color{BL9}
+$5107.0_Fill_Mode_Color{BL8}
BL32
BL31
BL28
BL37
BL38
BK44
BK41
BK43
BL36
BL46
BL45
BL41
BL42
BL39
BH25
BH27
BH22
ExRAM_D7_Read{BJ12(h)}
BH21
BH4
+PPU_Fetch_Counter.3{S21}
+V-Split_Scanline_Counter.4{T22}
BH24
ExRAM_D5_Read{BJ12(f)}
BH23
BH5
BH20
ExRAM_D3_Read{BJ12(d)}
BH19
BH6
BH26
BH18
ExRAM_D1_Read{BJ12(b)}
BH17
BH7
BH3
BH2
BH1
BH8
+ExRAM_Read_D7_Latch{BJ11}
ExRAM_D6_Read{BJ12(g)}
BH12
ExRAM_D4_Read{BJ12(e)}
BH13
ExRAM_D2_Read{BJ12(c)}
BH14
ExRAM_D0_Read{BJ12(a)}
BH15
BH16
+ExRAM_Read_D6_Latch{BJ11}
BH11
BH10
BH9
ExRAM_D0_Read{BJ12(a)}
BK10
BK9
PPU_D0_OUT_PIN
BK7
BK8
BK1
BL12
+$5106.0_Fill_Mode_Tile{L11}
BK14
BK13
PPU_D1_OUT_PIN
ExRAM_D1_Read{BJ12(b)}
BK12
BK11
BL1
BL13
+$5106.1_Fill_Mode_Tile{L12}
BK18
BK17
PPU_D2_OUT_PIN
ExRAM_D2_Read{BJ12(c)}
BK16
BK15
BL2
BL14
+$5106.2_Fill_Mode_Tile{L14}
BK22
BK21
PPU_D3_OUT_PIN
ExRAM_D3_Read{BJ12(d)}
BK19
BK20
BL3
BL15
+$5106.3_Fill_Mode_Tile{L13}
BK26
BK25
PPU_D4_OUT_PIN
ExRAM_D4_Read{BJ12(e)}
BK23
BK24
BL4
BL16
+$5106.4_Fill_Mode_Tile{L18}
BK30
BK29
PPU_D5_OUT_PIN
ExRAM_D5_Read{BJ12(f)}
BK27
BK28
BL5
BL17
+$5106.5_Fill_Mode_Tile{L17}
BK34
BK33
PPU_D6_OUT_PIN
ExRAM_D6_Read{BJ12(g)}
BK31
BK32
BL6
BL18
+$5106.6_Fill_Mode_Tile{L16}
BK38
BK37
PPU_D7_OUT_PIN
ExRAM_D7_Read{BJ12(h)}
BK35
BK36
BL7
BL19
+$5106.7_Fill_Mode_Tile{L15}
CIRAM_CE_PIN{A9}
BK2
BK5
BK4
+PPU_RD_A{BJ1}
+PPU_A13_B{BK6}
/PPU_A12_B{BK3}
PPU_Dx_Direction{BK4}
Sheet: /PPU Data Bus/
File: ppu_data_bus.kicad_sch
Title:
Size: USLedger
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 20/26

10-Bit CHR Bank Register Write Data Source Multiplexing

CPU_D0_IN_PIN
+$5101=%0x=8kB_or_4kB_Pages{AF3}
/$5101=%00=8kB_Pages{AE11}
?
Why did they not use a single tribuf 8N instead of AF5 + AF6?
AG1 INVERT
AF4 Must be +OE here
AF5 /OE TriBuf 1N
AF6 /OE TriBuf 7N
CPU_D0_IN_PIN
CPU_D1_IN_PIN
CPU_D2_IN_PIN
CPU_D3_IN_PIN
CPU_D4_IN_PIN
CPU_D5_IN_PIN
CPU_D6_IN_PIN
Note: CPU Dx Pins are non-inverted.

/$5101=%01=4kB_Pages{AE12}
AG2 /OE TriBuf 8N
CPU_D0_IN_PIN
CPU_D1_IN_PIN
CPU_D2_IN_PIN
CPU_D3_IN_PIN
CPU_D4_IN_PIN
CPU_D5_IN_PIN
CPU_D6_IN_PIN
CPU_D7_IN_PIN

/$5101=%10=2kB_Pages{AE13}
AH1 /OE TriBuf 9N
CPU_D0_IN_PIN
CPU_D1_IN_PIN
CPU_D2_IN_PIN
CPU_D3_IN_PIN
CPU_D4_IN_PIN
CPU_D5_IN_PIN
CPU_D6_IN_PIN
CPU_D7_IN_PIN
+$5130.0{AE17}

/$5101=%11=1kB_Pages{AE14}
AJ1 /OE TriBuf 9N
CPU_D1_IN_PIN
CPU_D2_IN_PIN
CPU_D3_IN_PIN
CPU_D4_IN_PIN
CPU_D5_IN_PIN
CPU_D6_IN_PIN
CPU_D7_IN_PIN
+$5130.0{AE17}
+$5130.1{AE17}

Note: The 10-bit CHR Reg Write bus appears to be inverted.

/CHR_REG_WRITE_D0
/CHR_REG_WRITE_D1
/CHR_REG_WRITE_D2
/CHR_REG_WRITE_D3
/CHR_REG_WRITE_D4
/CHR_REG_WRITE_D5
/CHR_REG_WRITE_D6
/CHR_REG_WRITE_D7
/CHR_REG_WRITE_D8
/CHR_REG_WRITE_D9

$5202 Register
V-Split CHR Bank

/CPU_D0{AE18}
/CPU_D1{AE19}
/CPU_D2{AE20}
/CPU_D3{AE25}
/CPU_D4{AE24}
/CPU_D5{AE23}
/CPU_D6{AE22}
/CPU_D7{AE21}
/CPU_$5202_WR{L82}
AE26 Reg 8 D-Latch
AE27 INVERT
AE28 INVERT
AE29 INVERT
AE19 INVERT
AE20 INVERT
AF21 INVERT
AF22 INVERT
AE30 INVERT
+VSplit_CHR_Bank_$5202.0{AE27}
+VSplit_CHR_Bank_$5202.1{AE28}
+VSplit_CHR_Bank_$5202.2{AE29}
+VSplit_CHR_Bank_$5202.3{AF19}
+VSplit_CHR_Bank_$5202.4{AF20}
+VSplit_CHR_Bank_$5202.5{AF21}
+VSplit_CHR_Bank_$5202.6{AF22}
+VSplit_CHR_Bank_$5202.7{AE30}

CHR Reg Write Buffering

/CHR_REG_WRITE_D0
AK5 BUFFER /CHR_REG_WRITE_D0{AK5}
AM5 BUFFER /CHR_REG_WRITE_D0{AM5}
AP5 BUFFER /CHR_REG_WRITE_D0{AP5}
BG1 BUFFER /CHR_REG_WRITE_D0{BG1}

/CHR_REG_WRITE_D1
AK4 BUFFER /CHR_REG_WRITE_D1{AK4}
AM4 BUFFER /CHR_REG_WRITE_D1{AM4}
AP4 BUFFER /CHR_REG_WRITE_D1{AP4}
BG2 BUFFER /CHR_REG_WRITE_D1{BG2}

/CHR_REG_WRITE_D2
AK3 BUFFER /CHR_REG_WRITE_D2{AK3}
AM3 BUFFER /CHR_REG_WRITE_D2{AM3}
AP3 BUFFER /CHR_REG_WRITE_D2{AP3}
BG3 BUFFER /CHR_REG_WRITE_D2{BG3}

/CHR_REG_WRITE_D3
AK2 BUFFER /CHR_REG_WRITE_D3{AK2}
AM2 BUFFER /CHR_REG_WRITE_D3{AM2}
AP2 BUFFER /CHR_REG_WRITE_D3{AP2}
BG4 BUFFER /CHR_REG_WRITE_D3{BG4}

/CHR_REG_WRITE_D4
AK1 BUFFER /CHR_REG_WRITE_D4{AK1}
AM1 BUFFER /CHR_REG_WRITE_D4{AM1}
AP1 BUFFER /CHR_REG_WRITE_D4{AP1}
BG5 BUFFER /CHR_REG_WRITE_D4{BG5}

/CHR_REG_WRITE_D5
AL5 BUFFER /CHR_REG_WRITE_D5{AL5}
AN5 BUFFER /CHR_REG_WRITE_D5{AN5}
AR5 BUFFER /CHR_REG_WRITE_D5{AR5}
BG6 BUFFER /CHR_REG_WRITE_D5{BG6}

/CHR_REG_WRITE_D6
AL4 BUFFER /CHR_REG_WRITE_D6{AL4}
AN4 BUFFER /CHR_REG_WRITE_D6{AN4}
AR4 BUFFER /CHR_REG_WRITE_D6{AR4}
BG7 BUFFER /CHR_REG_WRITE_D6{BG7}

/CHR_REG_WRITE_D7
AL3 BUFFER /CHR_REG_WRITE_D7{AL3}
AN3 BUFFER /CHR_REG_WRITE_D7{AN3}
AR3 BUFFER /CHR_REG_WRITE_D7{AR3}
BG8 BUFFER /CHR_REG_WRITE_D7{BG8}

/CHR_REG_WRITE_D8
AL2 BUFFER /CHR_REG_WRITE_D8{AL2}
AN2 BUFFER /CHR_REG_WRITE_D8{AN2}
AR2 BUFFER /CHR_REG_WRITE_D8{AR2}
BG9 BUFFER /CHR_REG_WRITE_D8{BG9}

/CHR_REG_WRITE_D9
AL1 BUFFER /CHR_REG_WRITE_D9{AL1}
AN1 BUFFER /CHR_REG_WRITE_D9{AN1}
AR1 BUFFER /CHR_REG_WRITE_D9{AR1}
BG10 BUFFER /CHR_REG_WRITE_D9{BG10}

$5130 Register
CHR Bank Upper Bits

/CPU_D0{AE16}
/CPU_D1{AE15}
/Reset_Detect{AE3}
/CPU_$5130_WR{L76}
AE17 /D0 /D1 /Reset Reg 2 D-Latch-Reset
/$5130.0{AE17}
+$5130.0{AE17}
/$5130.1{AE17}
+$5130.1{AE17}

$5120 Register

/CHR_REG_WRITE_D0
/CHR_REG_WRITE_D1
/CHR_REG_WRITE_D2
/CHR_REG_WRITE_D3
/CHR_REG_WRITE_D4
/CHR_REG_WRITE_D5
/CHR_REG_WRITE_D6
/CHR_REG_WRITE_D7
/CHR_REG_WRITE_D8
/CHR_REG_WRITE_D9
/CPU_$5120_WR{L56}
AF7 Reg 10 D-Latch N
$5120.0{AF7}
$5120.1{AF7}
$5120.2{AF7}
$5120.3{AF7}
$5120.4{AF7}
$5120.5{AF7}
$5120.6{AF7}
$5120.7{AF7}
$5120.8{AF7}
$5120.9{AF7}

$5121 Register

/CHR_REG_WRITE_D0
/CHR_REG_WRITE_D1
/CHR_REG_WRITE_D2
/CHR_REG_WRITE_D3
/CHR_REG_WRITE_D4
/CHR_REG_WRITE_D5
/CHR_REG_WRITE_D6
/CHR_REG_WRITE_D7
/CHR_REG_WRITE_D8
/CHR_REG_WRITE_D9
/CPU_$5121_WR{L58}
AG3 Reg 10 D-Latch N
$5121.0{AG3}
$5121.1{AG3}
$5121.2{AG3}
$5121.3{AG3}
$5121.4{AG3}
$5121.5{AG3}
$5121.6{AG3}
$5121.7{AG3}
$5121.8{AG3}
$5121.9{AG3}

$5122 Register

/CHR_REG_WRITE_D0
/CHR_REG_WRITE_D1
/CHR_REG_WRITE_D2
/CHR_REG_WRITE_D3
/CHR_REG_WRITE_D4
/CHR_REG_WRITE_D5
/CHR_REG_WRITE_D6
/CHR_REG_WRITE_D7
/CHR_REG_WRITE_D8
/CHR_REG_WRITE_D9
/CPU_$5122_WR{L62}
AH2 Reg 10 D-Latch N
$5122.0{AH2}
$5122.1{AH2}
$5122.2{AH2}
$5122.3{AH2}
$5122.4{AH2}
$5122.5{AH2}
$5122.6{AH2}
$5122.7{AH2}
$5122.8{AH2}
$5122.9{AH2}

$5123 Register

/CHR_REG_WRITE_D0
/CHR_REG_WRITE_D1
/CHR_REG_WRITE_D2
/CHR_REG_WRITE_D3
/CHR_REG_WRITE_D4
/CHR_REG_WRITE_D5
/CHR_REG_WRITE_D6
/CHR_REG_WRITE_D7
/CHR_REG_WRITE_D8
/CHR_REG_WRITE_D9
/CPU_$5123_WR{L64}
AJ2 Reg 10 D-Latch N
$5123.0{AJ2}
$5123.1{AJ2}
$5123.2{AJ2}
$5123.3{AJ2}
$5123.4{AJ2}
$5123.5{AJ2}
$5123.6{AJ2}
$5123.7{AJ2}
$5123.8{AJ2}
$5123.9{AJ2}

$5124 Register

8x16_Mode_Related{AS25}
Note: Reg 10 might have +OE input and /OE output, not sure.
/CHR_REG_WRITE_D0{AK5}
/CHR_REG_WRITE_D1{AK4}
/CHR_REG_WRITE_D2{AK3}
/CHR_REG_WRITE_D3{AK2}
/CHR_REG_WRITE_D4{AK1}
/CHR_REG_WRITE_D5{AL5}
/CHR_REG_WRITE_D6{AL4}
/CHR_REG_WRITE_D7{AL3}
/CHR_REG_WRITE_D8{AL2}
/CHR_REG_WRITE_D9{AL1}
/CPU_$5124_WR{L66}
AL6 Reg 10 D-Latch OE
$5124/8.0{AL6,AK6}
$5124/8.1{AL6,AK6}
$5124/8.2{AL6,AK6}
$5124/8.3{AL6,AK6}
$5124/8.4{AL6,AK6}
$5124/8.5{AL6,AK6}
$5124/8.6{AL6,AK6}
$5124/8.7{AL6,AK6}
$5124/8.8{AL6,AK6}
$5124/8.9{AL6,AK6}

$5125 Register

/CHR_REG_WRITE_D0{AM5}
/CHR_REG_WRITE_D1{AM4}
/CHR_REG_WRITE_D2{AM3}
/CHR_REG_WRITE_D3{AM2}
/CHR_REG_WRITE_D4{AM1}
/CHR_REG_WRITE_D5{AN5}
/CHR_REG_WRITE_D6{AN4}
/CHR_REG_WRITE_D7{AN3}
/CHR_REG_WRITE_D8{AN2}
/CHR_REG_WRITE_D9{AN1}
/CPU_$5125_WR{L69}
AN6 Reg 10 D-Latch OE
$5125/9.0{AN6,AM6}
$5125/9.1{AN6,AM6}
$5125/9.2{AN6,AM6}
$5125/9.3{AN6,AM6}
$5125/9.4{AN6,AM6}
$5125/9.5{AN6,AM6}
$5125/9.6{AN6,AM6}
$5125/9.7{AN6,AM6}
$5125/9.8{AN6,AM6}
$5125/9.9{AN6,AM6}

$5126 Register

/CHR_REG_WRITE_D0{AP5}
/CHR_REG_WRITE_D1{AP4}
/CHR_REG_WRITE_D2{AP3}
/CHR_REG_WRITE_D3{AP2}
/CHR_REG_WRITE_D4{AP1}
/CHR_REG_WRITE_D5{AR5}
/CHR_REG_WRITE_D6{AR4}
/CHR_REG_WRITE_D7{AR3}
/CHR_REG_WRITE_D8{AR2}
/CHR_REG_WRITE_D9{AR1}
/CPU_$5126_WR{L70}
AR6 Reg 10 D-Latch OE
$5126/A.0{AR6,AP6}
$5126/A.1{AR6,AP6}
$5126/A.2{AR6,AP6}
$5126/A.3{AR6,AP6}
$5126/A.4{AR6,AP6}
$5126/A.5{AR6,AP6}
$5126/A.6{AR6,AP6}
$5126/A.7{AR6,AP6}
$5126/A.8{AR6,AP6}
$5126/A.9{AR6,AP6}

$5127 Register

/CHR_REG_WRITE_D0{BG1}
/CHR_REG_WRITE_D1{BG2}
/CHR_REG_WRITE_D2{BG3}
/CHR_REG_WRITE_D3{BG4}
/CHR_REG_WRITE_D4{BG5}
/CHR_REG_WRITE_D5{BG6}
/CHR_REG_WRITE_D6{BG7}
/CHR_REG_WRITE_D7{BG8}
/CHR_REG_WRITE_D8{BG9}
/CHR_REG_WRITE_D9{BG10}
/CPU_$5127_WR{L74}
BG11 Reg 10 D-Latch OE
$5127/B.0{BG11,BF1}
$5127/B.1{BG11,BF1}
$5127/B.2{BG11,BF1}
$5127/B.3{BG11,BF1}
$5127/B.4{BG11,BF1}
$5127/B.5{BG11,BF1}
$5127/B.6{BG11,BF1}
$5127/B.7{BG11,BF1}
$5127/B.8{BG11,BF1}
$5127/B.9{BG11,BF1}

$5128 Register

/CHR_REG_WRITE_D0{AK5}
/CHR_REG_WRITE_D1{AK4}
/CHR_REG_WRITE_D2{AK3}
/CHR_REG_WRITE_D3{AK2}
/CHR_REG_WRITE_D4{AK1}
/CHR_REG_WRITE_D5{AL5}
/CHR_REG_WRITE_D6{AL4}
/CHR_REG_WRITE_D7{AL3}
/CHR_REG_WRITE_D8{AL2}
/CHR_REG_WRITE_D9{AL1}
/CPU_$5128_WR{L98}
AK6 Reg 10 D-Latch OE

$5129 Register

/CHR_REG_WRITE_D0{AM5}
/CHR_REG_WRITE_D1{AM4}
/CHR_REG_WRITE_D2{AM3}
/CHR_REG_WRITE_D3{AM2}
/CHR_REG_WRITE_D4{AM1}
/CHR_REG_WRITE_D5{AN5}
/CHR_REG_WRITE_D6{AN4}
/CHR_REG_WRITE_D7{AN3}
/CHR_REG_WRITE_D8{AN2}
/CHR_REG_WRITE_D9{AN1}
/CPU_$5129_WR{L99}
AM6 Reg 10 D-Latch OE

$512A Register

/CHR_REG_WRITE_D0{AP5}
/CHR_REG_WRITE_D1{AP4}
/CHR_REG_WRITE_D2{AP3}
/CHR_REG_WRITE_D3{AP2}
/CHR_REG_WRITE_D4{AP1}
/CHR_REG_WRITE_D5{AR5}
/CHR_REG_WRITE_D6{AR4}
/CHR_REG_WRITE_D7{AR3}
/CHR_REG_WRITE_D8{AR2}
/CHR_REG_WRITE_D9{AR1}
/CPU_$512A_WR{L102}
AP6 Reg 10 D-Latch OE

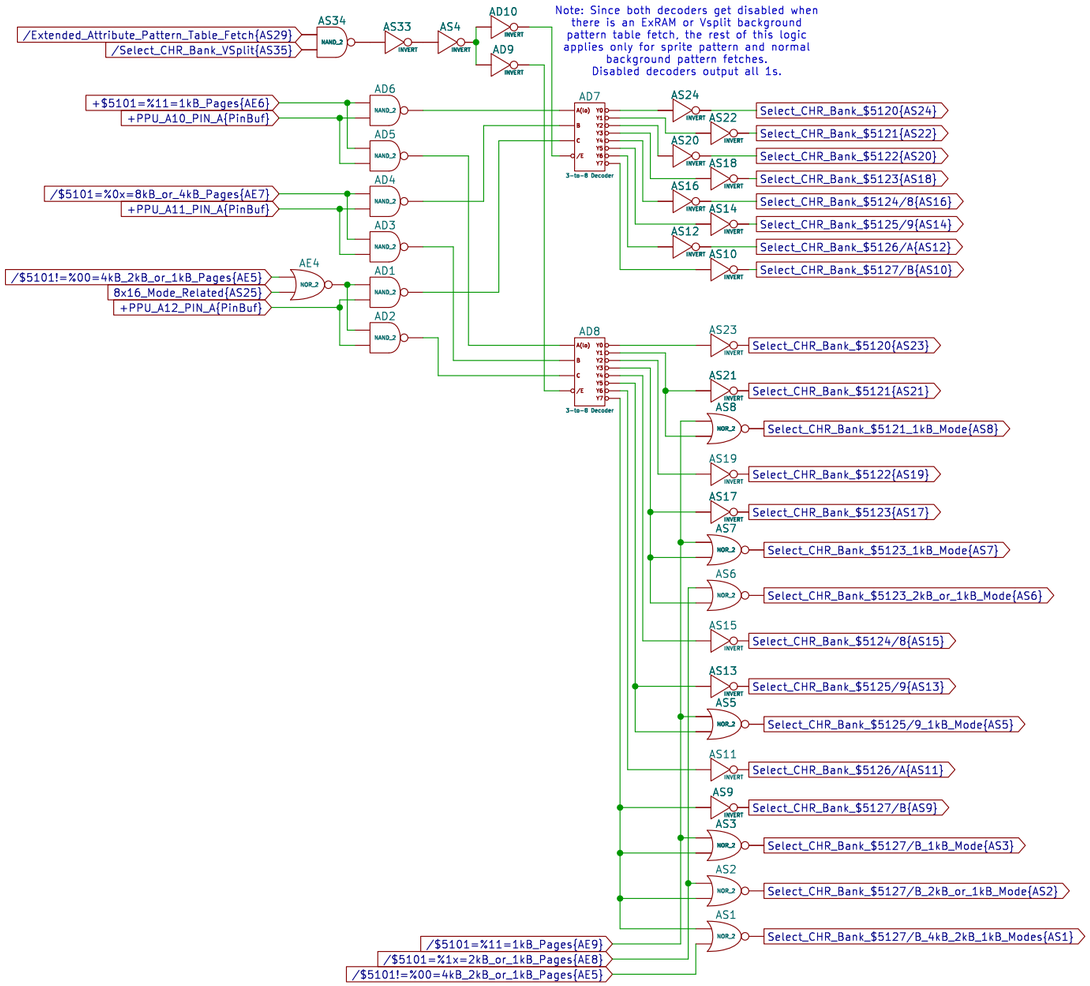
$512B Register

/CHR_REG_WRITE_D0{BG1}
/CHR_REG_WRITE_D1{BG2}
/CHR_REG_WRITE_D2{BG3}
/CHR_REG_WRITE_D3{BG4}
/CHR_REG_WRITE_D4{BG5}
/CHR_REG_WRITE_D5{BG6}
/CHR_REG_WRITE_D6{BG7}
/CHR_REG_WRITE_D7{BG8}
/CHR_REG_WRITE_D8{BG9}
/CHR_REG_WRITE_D9{BG10}
/CPU_$512B_WR{L104}
BF1 Reg 10 D-Latch OE

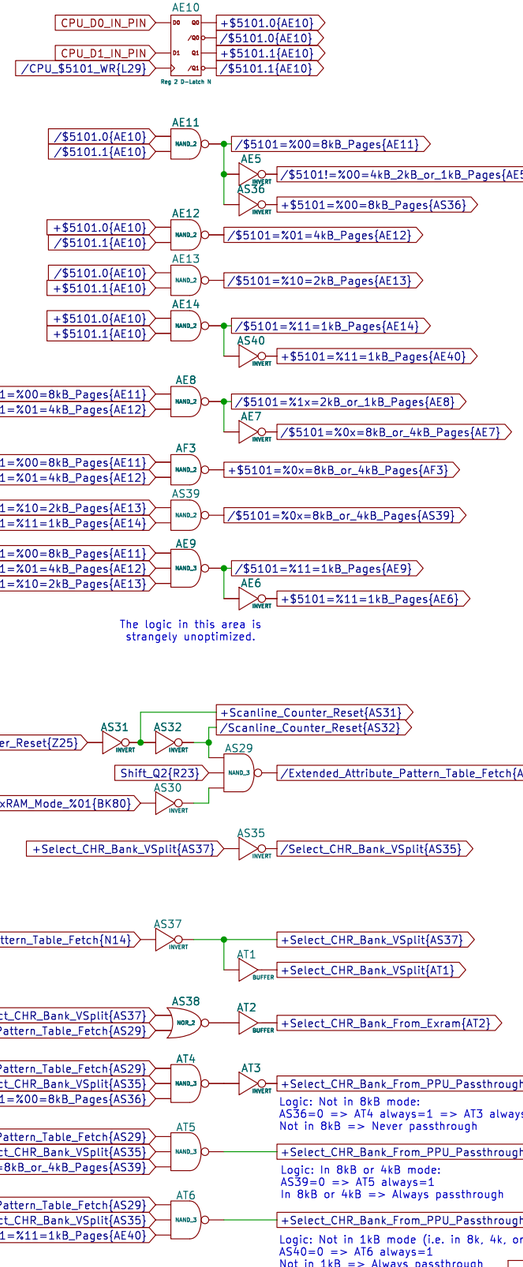
Sheet: /CHR Bank Registers/
File: chr_bank_registers.kicad_sch

Title:

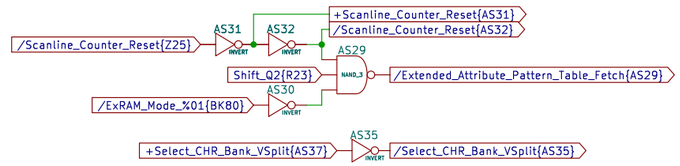
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 21/26

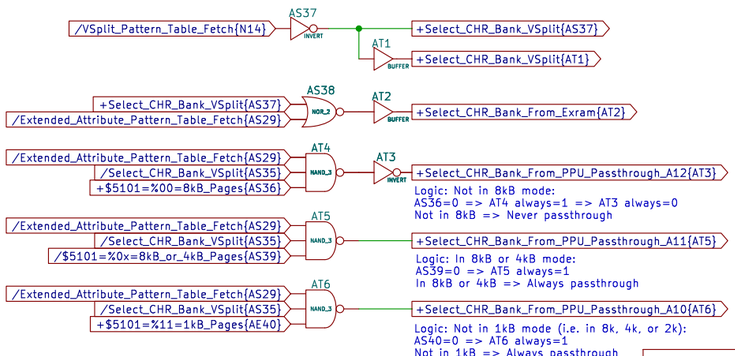


$5101 Register
CHR Mode

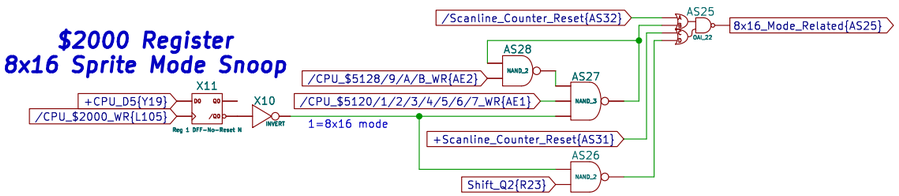


The logic in this area is strangely unoptimized.





$2000 Register
8x16 Sprite Mode Snoop

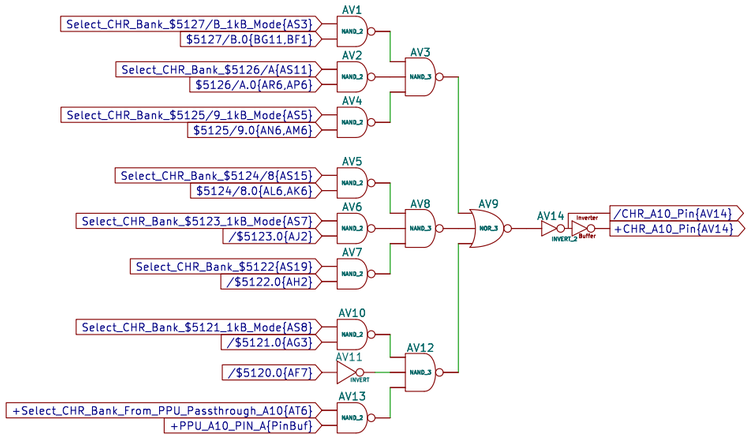


Sheet: /CHR Bank Selection Logic/
File: chr_bank_selection_logic.kicad_sch

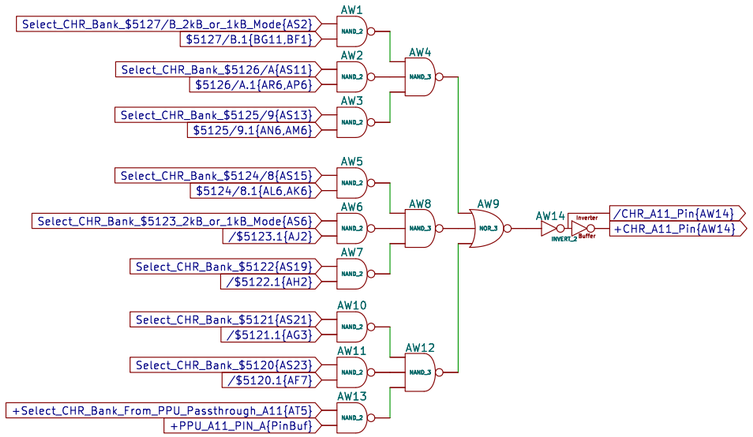
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Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 22/26

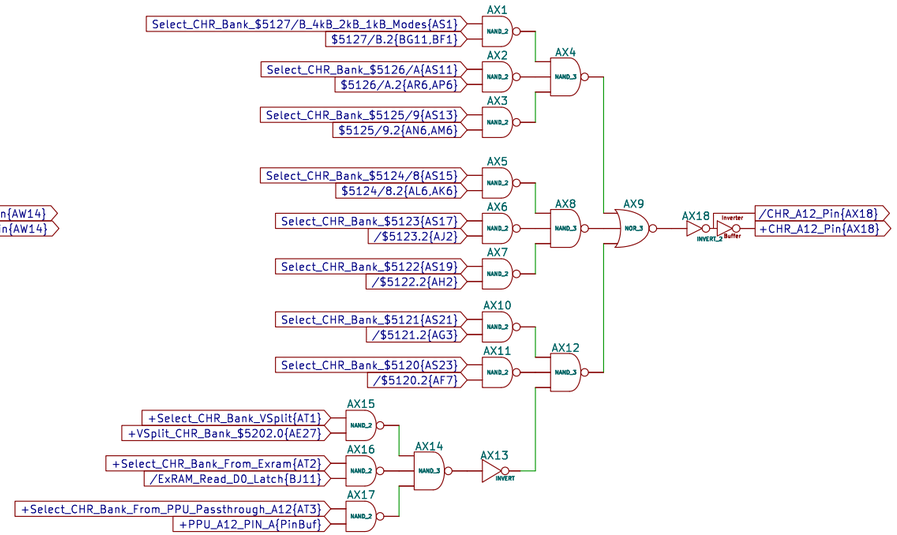
CHR A10



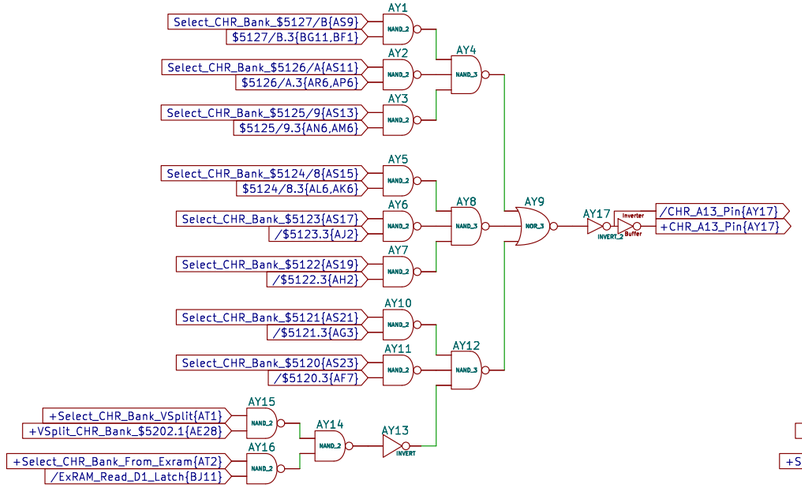
CHR A11



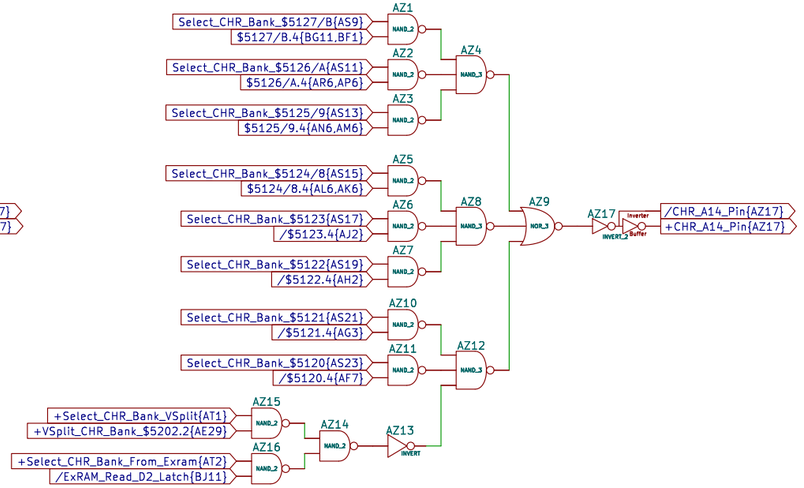
CHR A12



CHR A13



CHR A14

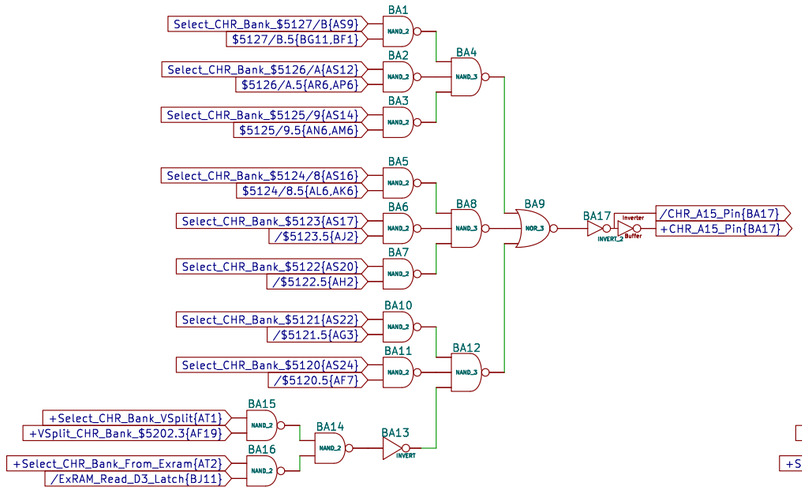


Sheet: /CHR Address Bus 10 to 14/
File: chr_address_bus_10_to_14.kicad_sch

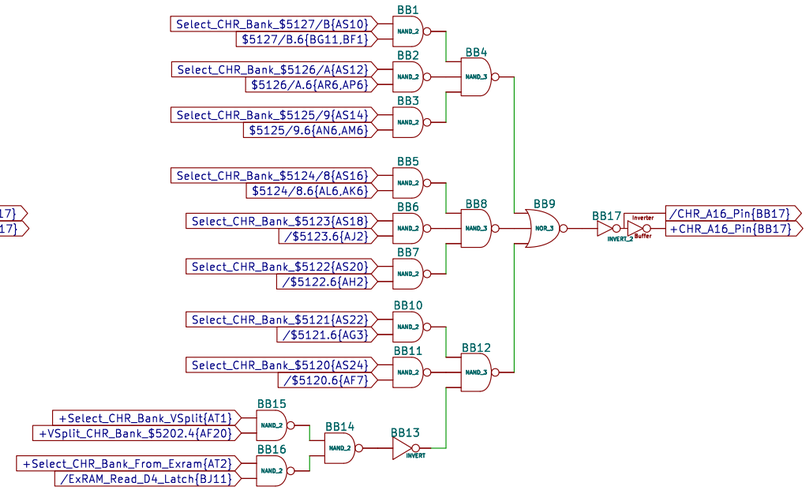
Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 23/26

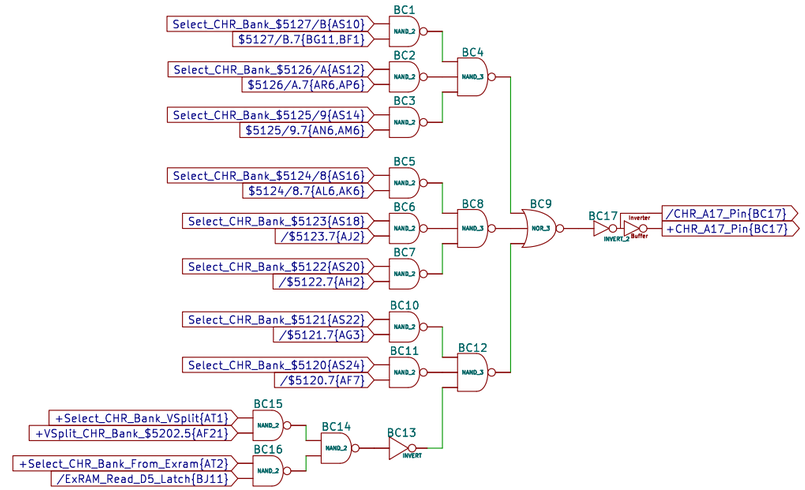
CHR A15



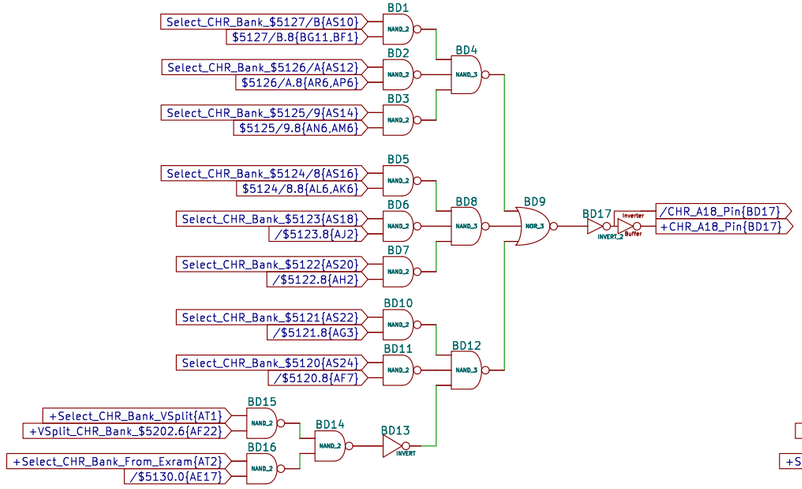
CHR A16



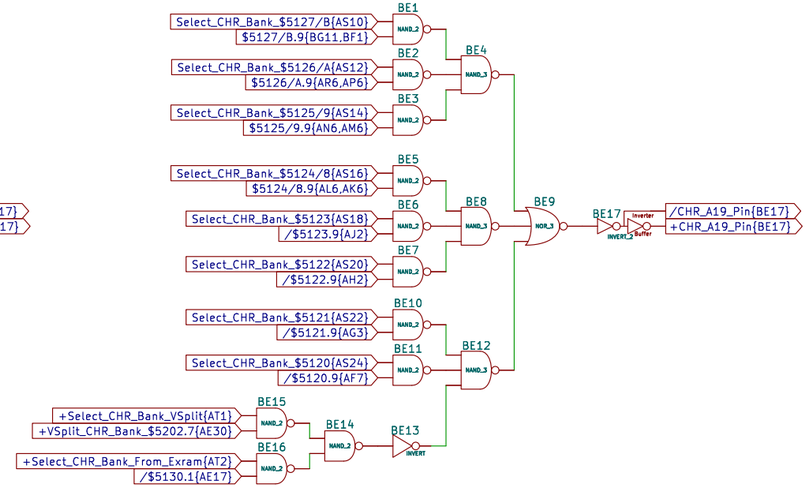
CHR A17



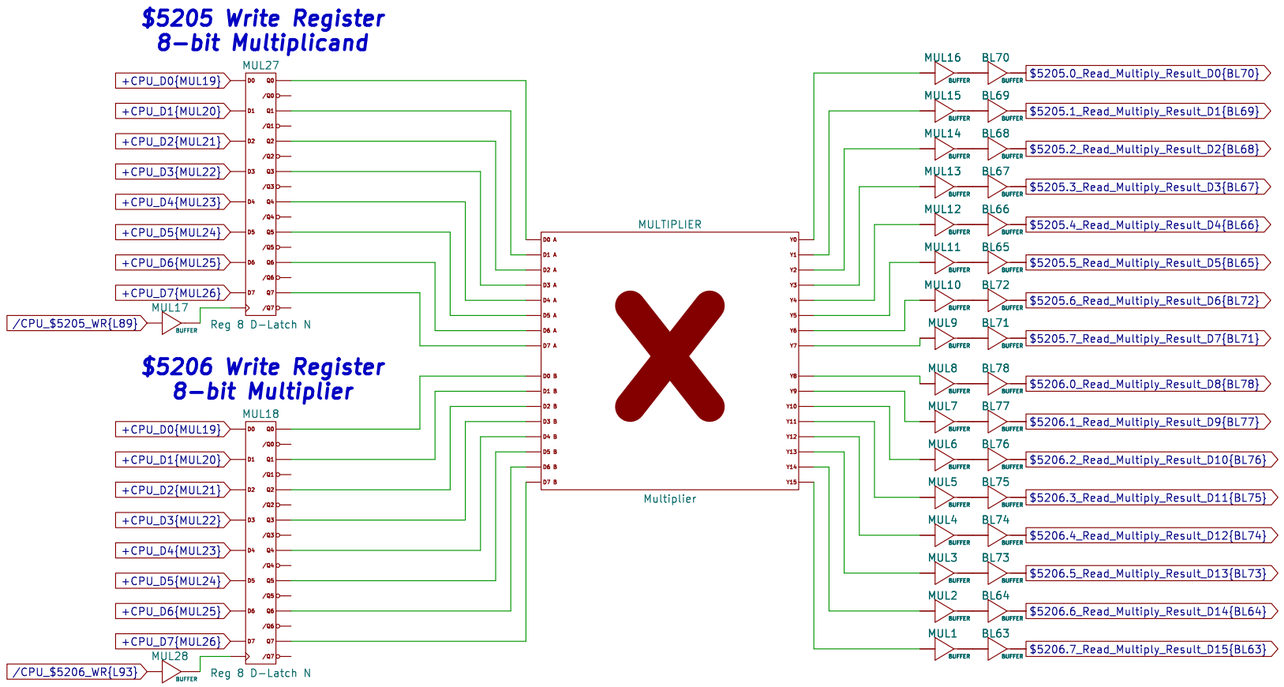
CHR A18



CHR A19



Sheet: /CHR Address Bus 15 to 19/	
File: chr_address_bus_15_to_19.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	**Rev:** Id: 24/26

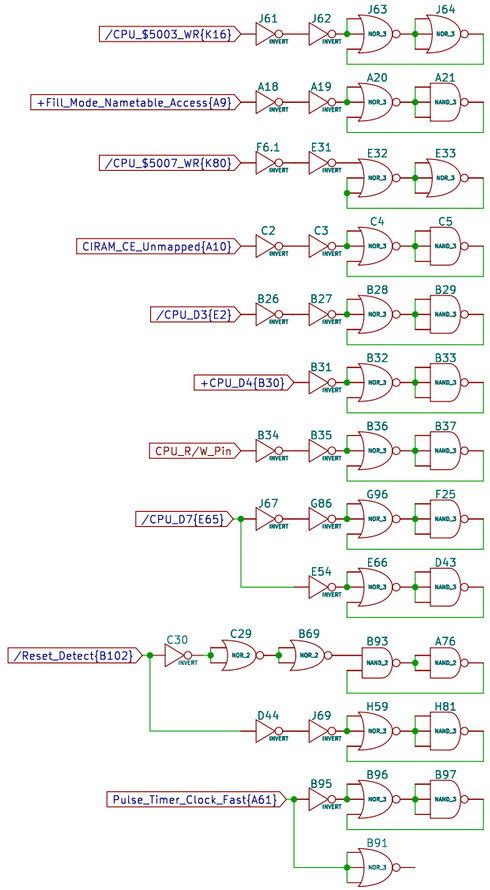


Sheet: /Multiplier/
File: multiplier.kicad_sch

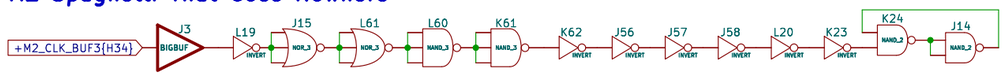
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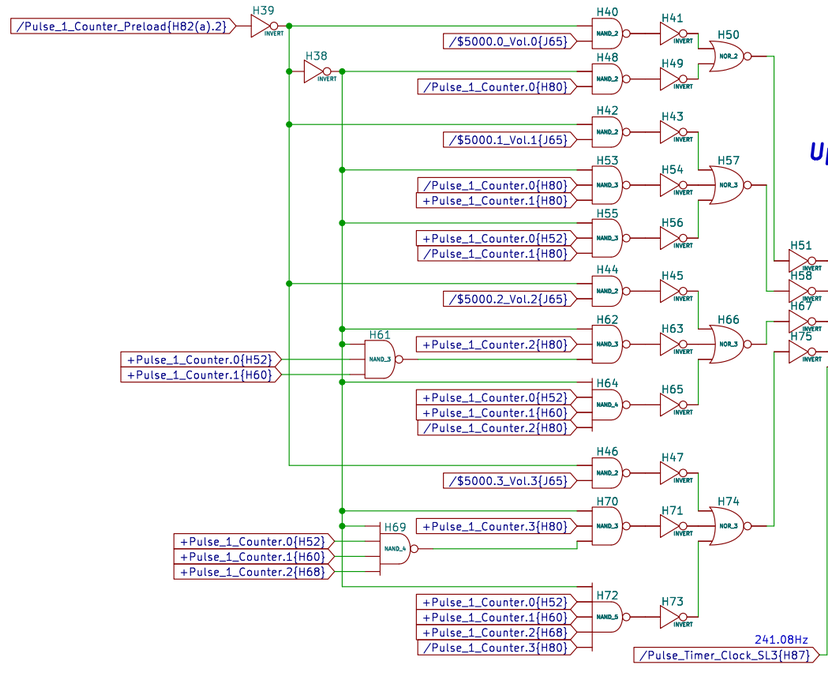
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 25/26

This sheet contains all residual items that don't serve any apparent purpose.



M2 Spaghetti That Goes Nowhere

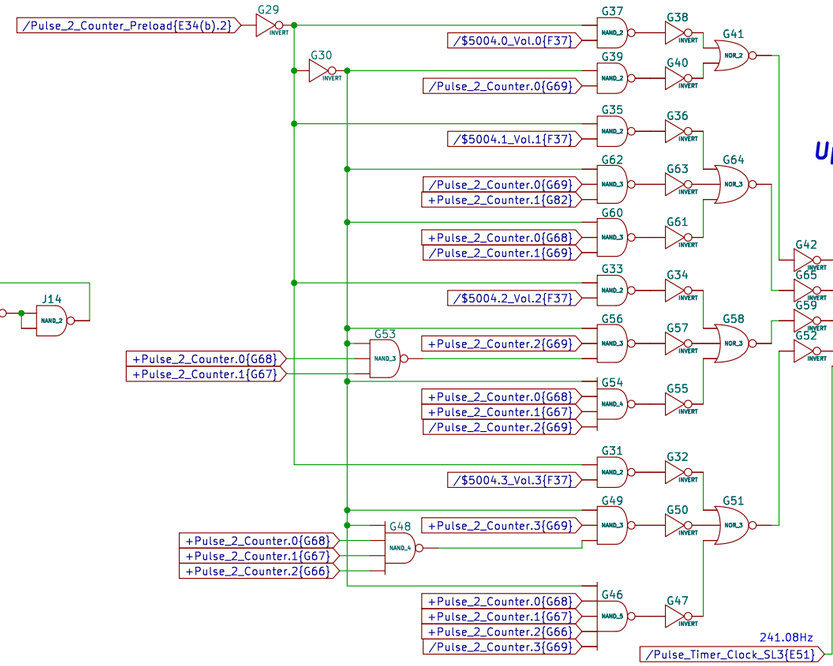




Pulse 1 Unused 4-bit Binary Up-Counter With Preload

The only output that goes elsewhere is H78, which gets latched by H82(a), coming directly back here as /Pulse_1_Counter_Preload; all self-contained. It does not appear to provide any other external signals, so I don't think it actually does anything. It seems like an abandoned envelope counter that was left in.

It latches the 4-bit pulse volume, then counts up at a rate of 240 Hz. Nothing stops it from counting or rolling over from $F back to $0.



Pulse 2 Unused 4-bit Binary Up-Counter With Preload

Sheet: /Artifacts/		
File: artifacts.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 26/26